Brain-age research summary: cohorts, experiments, the best model and what mattered

📄 **PDF version of this notebook** (figures, tables and formulas, no code; regenerate with `scripts/notebook_to_pdf.py notebooks/brain_age_research_summary.ipynb`).

One notebook for the whole EEG brain-age study on TUH (2026-10-05 → 10-16), in three phases:

- **Phase 1, TUAB only.** About 50 fine-tunes of LaBraM-base on the TUH Abnormal corpus (TUAB), aimed at memorization: schedule, regularizers, mixup/EMA, loss shape, codebook regularization, heads, training from scratch (`docs/age_overfitting_experiments.md`).
- **Phase 2, full TUH.** The same recipe on cohorts drawn from the whole TUH EEG corpus (TUEG, about 60k recordings), plus data, input and head changes (`docs/data/tuh_cohort_eval_v1.csv`).
- **Phase 3, cross-validation.** 5-fold patient-grouped CV of the three strongest phase-2 configurations (`notebooks/tuh_cv_summary.ipynb`, `docs/data/cv5_eval_v2.csv`).

It answers four questions:

1. Which datasets and cohorts were used, and how they differ (Section 1).
2. Which experiments ran and which came out near the best (Section 2).
3. What exactly the best experiment does, step by step (Section 3).
4. What changed relative to LaBraM as published (Section 4), and which of those changes measurably moved the error (Section 5).

Contents. 0 Metrics and evaluation sets · 1 Data · 2 Experiments · 3 The best experiment · 4 Changes versus the LaBraM recipe · 5 Which changes mattered · 6 Conclusions.

Inputs.

- In git: the registry, the two CSVs, `docs/data/line_removal_ab.json`, the configs, `docs/data/brain_age_clearml_runs.csv` (each run's ClearML experiment) and `scripts/submit_tuh_cohorts.sh`. Every phase-2 run's configuration is rebuilt from `labram/eval/research_summary.py::PHASE2` with the same config loader the trainer uses. `tests/test_research_summary.py` checks that catalog against the submit script.
- Optional, on the TUH data volume (`LABRAM_TUH_ROOT`, default `/data/datasets/EEG-public/TUH_EEG`):
  - the cohort manifests behind Section 1's counts;

- the TUH recording index and NEDC's clinical sheet behind the TUEG specification (Tables 1.3–1.6);
- the per-recording CV predictions behind the bootstraps (`LABRAM_CV_EVAL`).

They stay out of git under the TUH data-use agreement. Without them those cells print a note and skip, and the outputs show only aggregate counts.

0. Metrics and evaluation sets

Every error is computed **per recording**. A recording's predicted age is the mean of its window predictions (Eq. (3.7)), in years. For a set S of recordings with true age y_i and predicted age $\hat{y}_i$:

$$\mathrm{MAE}(S) = \frac{1}{|S|} \sum_{i \in S} |\hat{y}_i - y_i| \tag{0.1}$$

$$R^2(S) = 1 - \frac{\sum_{i \in S} (\hat{y}_i - y_i)^2}{\sum_{i \in S} (y_i - \bar{y}_S)^2} \tag{0.2}$$

$$\mathrm{bias}(S) = \frac{1}{|S|} \sum_{i \in S} (\hat{y}_i - y_i) \tag{0.3}$$

where $\bar{y}_S$ is the mean true age of S (`labram/eval/cv_compare.py::metrics`). The effect of a change from a baseline run b to a variant run v is

$$\Delta\mathrm{MAE} = \mathrm{MAE}_v(S) - \mathrm{MAE}_b(S), \tag{0.4}$$

so **negative is better**. Section 5 defines when a Δ is larger than run-to-run noise.

All errors are before bias correction. Every MAE in this notebook scores the raw prediction $\hat{y}_i$. The only exception is Table 3.5, which compares raw and bias-corrected errors.

The **bias slope** β measures regression to the mean, the typical failure of an age model: old patients are predicted too young and young ones too old. It is the least-squares slope of the error $e_i = \hat{y}_i - y_i$ against the true age, with population moments over S (`labram/utils/regression_metrics.py` , `age_bias_slope`):

$$\beta(S) = \frac{\mathrm{Cov}_S(y, e)}{\mathrm{Var}_S(y)} \tag{0.5}$$

- $\beta = 0$: the error does not depend on age.
- $\beta = -1$: the prediction ignores age entirely; it is a constant.
- $\beta = -0.4$: a patient 10 years older than average is predicted about 4 years too young.

Evaluation sets.

- **TUAB test:** TUAB's official evaluation set, 274 recordings of 251 patients (149 normal, 125 abnormal). It is the same 274 recordings in every phase (checked in Section 1) and no training split contains any of its patients, so it is the one yardstick every run shares.
- **Val:** each run's own validation split. It differs between cohorts, so it compares runs only within a cohort.
- **CV test / TUEG out-of-fold:** each recording of the CV pool scored by the one fold model that never saw its patient.

Caveat. Phases 1 and 2 compared many configurations on the TUAB test set, which makes their best single-split numbers optimistically selected. The CV numbers (Phase 3) and the TUEG out-of-fold error were not used to choose anything.

```
registry: 45 phase-1 runs; cohort CSV: 19 phase-2 runs; CV CSV: 3 experiments x 5 folds
TUH data volume: found; per-recording CV predictions: found
```

1. Data: every dataset and cohort

All data comes from the TUH EEG corpus. TUAB (TUH Abnormal, about 3k recordings with a normal/abnormal label) is a subset of TUEG (the full TUH EEG corpus, about 60k recordings). The age label is parsed from each EDF header (`labram/data/tuh_metadata.py`). Every dataset uses the same signal format:

- 23 referential channels at 200 Hz in µV, 0.1–75 Hz band-pass;
- 10 s windows;
- the first and last 60 s of each recording dropped;
- evaluation on the first 5 minutes after the trim (`labram/data/window_selection.py`).

The datasets differ in which recordings they contain, how mains interference is removed, and how patients are split.

Lineage.

- `TUAB v3.0.0` (phase 1) → `TUAB v3.0.1 interp` (phase-2 control): same recordings and split, mains removed by 50 + 60 Hz spectrum interpolation instead of a 50 Hz notch.
- `normal_TUEG_v1` , `normal_both_v1` : normal recordings only (TUEG alone; TUAB + TUEG).
- `all_both_v1` → `v2` → `v3` : every recording (normal, abnormal, unlabelled), with TUAB patients kept in their TUAB split.
  - `v2` gives TUEG one patient split shared by every cohort, so validation is comparable across cohorts.
  - `v3` adds the TUEG recordings that lack only the T1/T2 electrodes.
- `v3_compact` / `v3_cvcompact` : byte-for-byte subsets of `v3` that store only the windows a run reads (`dataset_maker/make_compact_cohort.py`). `cvcompact` keeps both training and evaluation windows of every recording, so any CV fold can use any recording.

Tables 1.3–1.6 and Figure 1.2 at the end of this section specify TUEG itself: size, subsets, labels and age.

All cohorts are built by `dataset_maker/make_cohort_dataset.py`, which drops every TUEG recording of a TUAB-test patient and asserts that no patient is in two splits.

Table 1.1. Datasets and cohorts

One row per dataset. Names show the corpora in capitals (TUAB, TUEG); the directories on disk under `$LABRAM_TUH_ROOT/cohorts/` use the same names in lower case (e.g. `normal_tueg_v1`).

column	meaning
`phase`	the phase that used the dataset
`contents`	which recordings it holds
`line removal`	how mains interference is removed (from the dataset's `manifest.json`)
`rules`	inclusion rules: `age` range in years; `≥ min` minimum recording length in minutes; `≤ k/patient` at most *k* train/val recordings per patient; `T1/T2 may be missing`
`TUEG split`	how TUEG patients are split; TUAB patients always keep their TUAB split
`used by`	the runs of Tables 2.2–2.4 trained on it

dataset	phase	contents	line removal	rules	TUEG split	used by
TUAB v3.0.0	1	TUAB, all recordings (normal + abnormal)	notch:50	age 1–89 (TUAB)	TUAB: seeded 20 % val, official test	phase-1 runs (Table 2.2)
TUAB v3.0.1 interp	2	TUAB, all recordings	interp:50,60	age 1–89 (TUAB)	TUAB: seeded 20 % val, official test	Control
normal_TUEG_v1	2	TUEG normal, patients outside TUAB; no test split	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient	seeded 10 % val	Exp2
normal_both_v1	2	TUAB normal + TUEG normal	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient	seeded 10 % val	Exp1, Exp1b, Exp1t, Exp1f
all_both_v1	2	TUAB + TUEG, every recording (normal / abnormal / no label)	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient	seeded 10 % val	Exp3, Exp3b, Exp3q, Exp3c
all_both_v2	2	as v1, TUEG patients on the shared split	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient	shared patient split v2	Exp3v2, Exp3cond, Exp3clscond
all_both_v3	2	v2 + TUEG recordings missing only T1/T2	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient, T1/T2 may be missing	shared patient split v2	Exp4drop, Exp4base, Exp4ctrl, Exp4aug, Exp4aug15
all_both_v3_compact	2	v3, only the windows Exp4base reads	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient, T1/T2 may be missing	shared patient split v2	Exp4basec
all_both_v3_cvcompact	3	v3, training + evaluation windows of every recording (CV pool = train + val)	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient, T1/T2 may be missing	shared patient split v2	cvbase, cvctrl, cvaug

Table 1.2. Size and composition per split

For each dataset and split:

- `recordings` and `patients`;
- `from TUAB`: recordings that are also in TUAB;
- `normal` / `abnormal` / `no label`: the recording's report status (TUEG recordings without a normal/abnormal report have no label);
- `age mean` and `age sd`: the true age's mean and sample standard deviation (years).

Computed from each dataset's `cohort.parquet` / `manifest.json`, or from `age_split.json` + `age_metadata.json` for the TUAB-only datasets (`labram/eval/research_summary.py::cohort_summary`, `tuab_split_summary`).

dataset	split	recordings	patients	from TUAB	normal	abnormal	no label	age mean	age sd
TUAB v3.0.0	train	2145	1650	2145	1075	1070	0	49.2	17.7
	val	556	412	556	291	265	0	48.4	15.9
	test	274	251	274	149	125	0	50.4	18.1
TUAB v3.0.1 interp	train	2145	1650	2145	1075	1070	0	49.2	17.7
	val	556	412	556	291	265	0	48.4	15.9
	test	274	251	274	149	125	0	50.4	18.1
normal_TUEG_v1	train	3763	2150	0	3763	0	0	48.6	17.6
	val	395	239	0	395	0	0	49.9	16.9
normal_both_v1	train	5450	2995	775	5450	0	0	47.2	17.2
	val	843	458	196	843	0	0	46.6	15.7
	test	274	251	274	149	125	0	50.4	18.1
all_both_v1	train	20368	9069	1879	8204	10610	1554	52.0	17.5
	val	2905	1235	478	1254	1310	341	51.3	16.3
	test	274	251	274	149	125	0	50.4	18.1
all_both_v2	train	20470	9087	1879	8277	10617	1576	51.9	17.5
	val	2803	1217	478	1181	1303	319	51.6	16.5
	test	274	251	274	149	125	0	50.4	18.1
all_both_v3	train	25613	12017	1868	10067	13714	1832	52.4	17.5
	val	3422	1533	482	1388	1676	358	52.1	16.8
	test	274	251	274	149	125	0	50.4	18.1
all_both_v3_compact	train	25613	12017	1868	10067	13714	1832	52.4	17.5
	val	3422	1533	482	1388	1676	358	52.1	16.8
	test	274	251	274	149	125	0	50.4	18.1
all_both_v3_cvcompact	train	25613	12017	1868	10067	13714	1832	52.4	17.5
	val	3422	1533	482	1388	1676	358	52.1	16.8
	test	274	251	274	149	125	0	50.4	18.1

```
TUAB test: 274 recordings; identical in every dataset with a test split: True
{'TUAB v3.0.0': True, 'TUAB v3.0.1 interp': True, 'normal_both_v1': True, 'all_both_v
1': True, 'all_both_v2': True, 'all_both_v3': True, 'all_both_v3_compact': True, 'all_b
oth_v3_cvcompact': True}
```

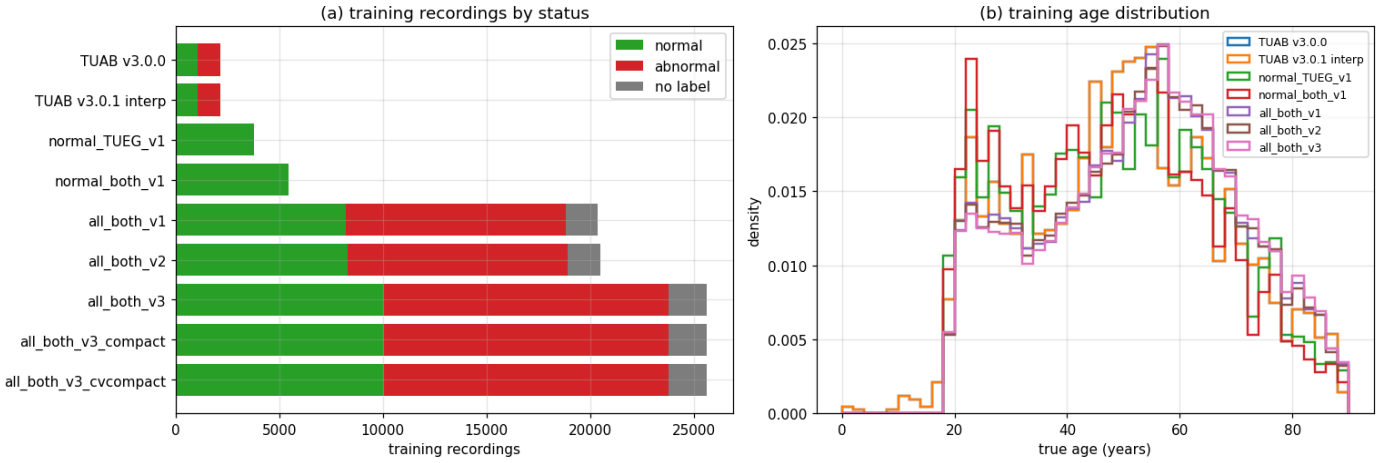
Reading Table 1.2.

- **Scale:** training grows from 2,145 TUAB recordings (phase 1) to 25,613 recordings of about 12k patients (`all_both_v3`).
- **Composition:** in the `all_both` cohorts about half the training recordings are abnormal and about 7 % have no label.
- **Normal-only cohorts:** `normal_*` hold only normal recordings, but their test split is still the full TUAB test set, abnormal recordings included.
- **Test set:** the 274 TUAB test recordings are identical across datasets (the check above), so every TUAB test number in Sections 2–5 is on the same recordings.

Figure 1.1. Composition and age of the training sets

- (a) Training recordings per dataset, stacked by status (normal / abnormal / no label).
- (b) Distribution of true age (years) in each dataset's training split, as a density (step histogram, 2-year bins). Datasets with identical training sets (the `v3` copies) are drawn once.

Data volume required.



TUEG, the source corpus. Every cohort above draws from TUEG v2.0.2, the full TUH EEG corpus (`$LABRAM_TUH_ROOT/tueg/v2.0.2`). Tables 1.3–1.6 and Figure 1.2 specify it from two files:

- the TUH recording index, one row per EDF over all TUH corpora (`index/tuh_index.parquet` , `dataset_maker/tuh_index.py`);
- NEDC's clinical sheet, one row per session, joined on patient + session (`index/nedc/metadata_v12.xlsm.xlsx` , `dataset_maker/tuh_clinical.py`).

`labram/eval/research_summary.py::tueg_recordings` builds the joined table. For a set S of the $N = |\mathrm{TUEG}|$ recordings, with recording i lasting d_i seconds:

$$H(S) = \frac{1}{3600} \sum_{i \in S} d_i, \qquad \%(S) = 100 \, \frac{|S|}{N} \tag{1.1}$$

A recording's **status** is the TUAB folder label where the recording is in TUAB. Otherwise it is the report finding of the clinical sheet: abnormal when the sheet's "abnormal due to" field is filled. Recordings whose session is not in the sheet have no label (`dataset_maker/make_cohort_dataset.py::label_of` , the label the `all_both` cohorts use):

$$\mathrm{status}_i = \begin{cases} \mathrm{TUAB\ label} & i \in \mathrm{TUAB} \\ \text{abnormal if the report gives a cause, else normal} & i \notin \mathrm{TUAB},\ \text{session} \\ \text{no label} & \text{otherwise} \end{cases}$$

Ages are in years, from the EDF header. TUH writes 999 for patients aged 90 or older (a HIPAA redaction), and those recordings count as age-unknown. Age statistics use the recordings with a known age a_i: mean $\bar{a}$, sample standard deviation and quartiles Q_1, median, Q_3:

$$\bar{a} = \frac{1}{n} \sum_{i=1}^{n} a_i, \qquad s_a = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (a_i - \bar{a})^2} \tag{1.3}$$

Data volume required.

Table 1.3. TUEG at a glance

One row per quantity (`rs.tueg_overview`):

- `size` : recordings, patients, sessions and hours (Eq. (1.1));
- `recording` : length, year, sampling rate and montage. The year range omits a few EDF headers with impossible dates;
- `channels` : whether the recording has all 23 channels the models read;
- `patient` : sex and age (Eq. (1.3)), counted over recordings, so a patient with several recordings counts several times.

group	quantity	value
size	recordings (EDF files)	70,830
	patients	14,987
	sessions	26,845
	hours of EEG	27,021
	recordings per patient, median (max)	1 (713)
recording	length (min), median [IQR]	10.0 [5.0, 20.3]
	year, median [5 %, 95 %]	2014 [2003, 2016]
	sampling rate (Hz)	256: 52%, 250: 37%, 1000: 4%
	montage	01_tcp_ar: 73%, 02_tcp_le: 18%, 03_tcp_ar_a: 9%
channels	all 23 channels	55,182 (77.9 %)
	missing only T1/T2	8,479 (12.0 %)
	missing other channels	7,169 (10.1 %)
patient	sex F / M / unknown (recordings)	36,519 / 33,785 / 526
	age known (1–89 y)	69,747 (98.5 %)
	age 90+ (redacted as 999)	714 (1.0 %)
	age missing	369 (0.5 %)
	age (y), mean ± sd; median [IQR]	51.0 ± 18.1; 53 [38, 64]

Table 1.4. Subsets of TUEG

The other TUH corpora are subsets of TUEG: the same EDF files, each corpus adding its own label. One recording can be in several of them, so the rows overlap (`rs.tueg_subsets`). The last four rows are the subsets the brain-age cohorts select from:

- `TUEG only` : in no other corpus;
- `in the NEDC clinical sheet` : has a report status (Eq. (1.2));
- `adult age (18–89 y)` : the age range of the cohorts;
- `normative` : the pool of the `normal_*` cohorts (`dataset_maker/tuh_clinical.py::normative_mask`).

Columns: `recordings` , `% of TUEG` and `hours` (Eq. (1.1)), and `patients` . TUEV is not listed: none of its recordings is in TUEG v2.0.2.

subset	label it carries	recordings	% of TUEG	patients	hours
in TUAB	normal / abnormal (TUAB folder)	2993	4.2	2329	1141.0
in TUSZ	seizure events	8140	11.5	675	1480.7
in TUEP	epilepsy / no epilepsy (patient)	2808	4.0	200	635.5
in TUAR	artifact events	310	0.4	213	95.2
in TUSL	slowing events	112	0.2	38	26.8
TUEG only	–	57836	81.7	13448	24024.5
in the NEDC clinical sheet	report finding, 15 diagnosis flags, medications	60272	85.1	14508	24956.2
adult age (18–89 y)	age	68077	96.1	14025	25795.6
normative	report-normal, no neurological diagnosis, not sedated, 18–89 y	14343	20.2	4773	4569.7

Table 1.5. TUEG status labels and age per label

One row per status (Eq. (1.2)) and label source (`rs.tueg_status`):

- `TUAB folder` : TUAB's normal/abnormal label;
- `NEDC report` : the clinical sheet's finding;
- `all` : both sources;
- `total` : all of TUEG.

Columns:

- `recordings` , `% of TUEG` (Eq. (1.1));
- `patients` : a patient whose recordings differ in status counts under each status, so the rows sum to more than the total;
- `age n` : recordings with a known age;
- `age mean` , `age sd` (Eq. (1.3)), and `age Q1` , `age median` , `age Q3` in years.

Where TUAB and the report both label a session, they agree on about 95 % of its recordings.

status	source	recordings	% of TUEG	patients	age n	age mean	age sd	age Q1	age median	age Q3
normal	**NEDC report**	25340	35.8	6403	24921	48.2	17.9	34.0	50.0	62.0
	TUAB folder	1521	2.1	1385	1515	44.0	16.1	31.0	44.0	55.0
	all	26861	37.9	7348	26436	47.9	17.8	34.0	49.0	62.0
abnormal	**NEDC report**	32373	45.7	8132	31855	53.3	18.1	41.0	56.0	67.0
	TUAB folder	1472	2.1	998	1463	54.4	17.2	44.0	55.0	67.0
	all	33845	47.8	8664	33318	53.4	18.0	41.0	56.0	67.0
no label	**–**	10124	14.3	808	9993	51.1	17.7	39.0	53.0	63.0
total	**all**	70830	100.0	14987	69747	51.0	18.1	38.0	53.0	64.0

Figure 1.2. TUEG age distribution per status and in total

True age (years) of the TUEG recordings with a known age ($n =$ `age n` of Table 1.5), in 2-year bins, per status (Eq. (1.2)) and for all of TUEG:

- (a) Recordings per bin, stacked by status. The stack's outline is the total.
- (b) The same distributions as densities (each integrates to 1), so their shapes compare despite the different group sizes. The vertical ticks mark each group's median.

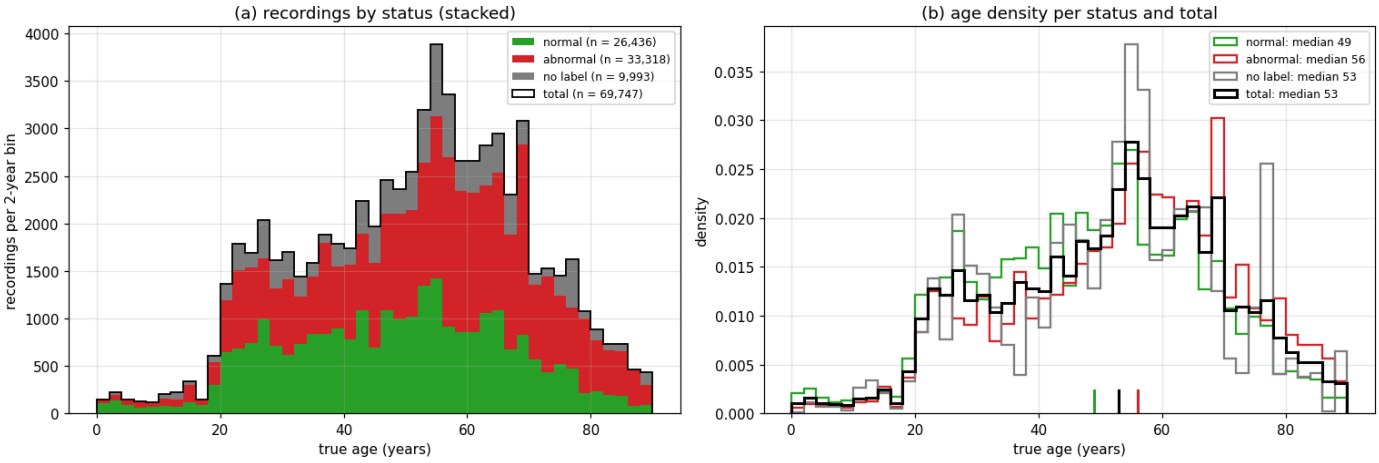


Table 1.6. TUEG diagnosis flags

The clinical sheet's 15 diagnosis flags and two derived flags over the TUEG recordings in the sheet, most frequent first (`rs.tueg_diagnoses`):

- `any neurological diagnosis` : seizure (clinical or electrographic), epilepsy, stroke, Parkinson's, Alzheimer's, TBI, dementia or MS;
- `sedating medication` : an anaesthetic, benzodiazepine, barbiturate or opioid on the medication list.

Columns:

- `recordings` and `patients` with the flag;
- `% of sheet` : Eq. (1.1) relative to the sheet's recordings instead of N;
- `age mean` : Eq. (1.3), in years;
- `% abnormal` : the share with status abnormal (Eq. (1.2)).

The 14 pathology groups of the condition vector (Eq. (3.6)) come from these flags.

flag	recordings	patients	% of sheet	age mean	% abnormal
sedating medication	23067	4727	38.3	52.3	60.4
any neurological diagnosis	11560	1736	19.2	51.1	66.3
Clinical seizure	6980	1092	11.6	48.5	58.1
Epilepsy	4660	306	7.7	47.0	57.3
Electrographic seizure	3015	66	5.0	54.5	89.1
Stroke	2590	539	4.3	61.3	84.4
Substance abuse / dependence	2083	725	3.5	45.7	71.6
TBI	720	202	1.2	48.6	79.6
Bipolar disorder	502	240	0.8	42.7	51.6
Dementia	293	50	0.5	72.3	78.8
Depression	213	77	0.4	44.7	35.7
Schizophrenia	191	43	0.3	52.5	31.9
Anxiety	182	72	0.3	40.5	39.6
Alzheimer's	166	6	0.3	68.3	98.2
Multiple sclerosis	108	37	0.2	48.0	46.3
ADHD	78	37	0.1	25.4	39.7
Parkinson's	9	3	0.0	55.4	22.2

Reading Tables 1.3–1.6 and Figure 1.2.

- **Size:** TUEG holds about 70.8k recordings (27k hours) of about 15k patients. The cohorts use 25k–29k of them, after the age, length, channel and per-patient rules of Table 1.1.
- **Recordings per patient:** most patients have a single recording, but a few long-monitored patients have hundreds. This is why the cohorts cap train/val recordings per patient and every split is by patient.
- **Labels:** only 4 % of TUEG is in TUAB. The status of nearly all other recordings comes from the NEDC report: about half are abnormal, and 14 % have no label (Table 1.5).
- **Age:** abnormal recordings are older than normal ones by about 5 years in mean and 7 in median (Table 1.5, Figure 1.2(b)). A model trained on mixed statuses can therefore partly read age from pathology, which is one reason Section 5 reports normal and abnormal test recordings separately.

2. Experiments

Phase 1 (TUAB only) starts from scenario **D**, the default `finetune_tuab_age.json` (Table 2.1 lists how it differs from LaBraM's own fine-tuning recipe):

- LaBraM-base with a **common average reference (CAR)** on the input (Eq. (3.1));
- Huber loss on the z-scored age;
- lr 10^{-4}, 15 epochs.

CAR re-references the EEG. At every time sample it subtracts the mean over all channels from each channel, so every channel is measured against the average of all electrodes instead of the recording's own reference. This removes what all electrodes share: the activity of the reference electrode(s) and noise common to every channel. It also puts the two TUH montages on one reference, average reference (`tcp_ar` , about 80 % of the recordings) and linked ears (`tcp_le`). LaBraM as published feeds the signal on each recording's own reference. CAR is the only part of the LaBraM++ input block D uses: there is no per-patch z-scoring (Table 2.1). In Table 5.2, row P1 tests it.

E2 shortens it to 4 epochs / 1 warmup epoch, and **M1** = E2 + mixup $\alpha = 0.4$. Every **phase-2** run is M1 plus the change named in its row (`labram/eval/research_summary.py::M1`). **Phase 3** cross-validates the three strongest phase-2 configurations.

Loss. Every run trains on a loss from the family of Eqs. (2.1)–(2.11) below, and Tables 2.2–2.4 name each run's loss in their `loss` column.

ClearML. Tables 2.2–2.5 link every run to its ClearML experiment; the link text is the experiment's name in ClearML (`docs/data/brain_age_clearml_runs.csv` , written from the ClearML server by `labram/eval/research_summary.py::fetch_clearml_runs`). A CV run links to the ClearML project that holds its five fold experiments.

Table 2.1. Base configuration D versus the LaBraM fine-tuning

recipe

Every configuration field on which LaBraM's TUAB fine-tuning recipe as published and the base configuration D differ:

- `LaBraM paper`: `labram/configs/defaults/finetune_tuab_paper.json`, the normal/abnormal fine-tune of LaBraM-base from the paper and the README;
- `D`: `labram/configs/defaults/finetune_tuab_age.json` as is, without overrides.

Both are rebuilt by `rs.resolve_config` and compared by `labram/eval/research_summary.py::config_diff`. Excluded:

- bookkeeping fields (paths, logging, infrastructure);
- classification-only fields a regression run never reads (`INERT_FOR_REGRESSION`, e.g. label smoothing);
- fields of a section that is off in both.

`group` is the part of the pipeline and `meaning` what the change does. Every other field is the paper's: LaBraM-base from `labram-base.pth`, batch 64, AdamW with weight decay 0.05, layer decay 0.65, drop path 0.1, mean pooling, cosine schedule. Changes that live in the data rather than the configuration (mains-interference removal, splits, labels) are in Table 4.2; how the best configuration differs from the paper's recipe is Table 4.1.

group	field	LaBraM paper	D	meaning
task	**data.dataset**	TUAB	TUAB_AGE	age regression on TUAB (age from the EDF header) instead of normal/abnormal
	model.task	classification	regression	scalar regression head, Huber loss on the z-scored age (Eqs. (3.2), (3.4))
data	**data.data_format**	pickle	npy	storage only: one float32 npy per recording, windows bit-identical to the pickles
input	**labram_plus.enabled**	False	True	LaBraM++ input block on: common average reference, CAR (Eq. (3.1))
	labram_plus.z_score_patches	True	False	LaBraM++ block with CAR only, no per-patch z-scoring
windows	**data.eval_minutes**	0	5	score each recording on its first 5 minutes after the trim
	data.trim_end_sec	0	60	drop the last 60 s of every recording
	data.trim_start_sec	0	60	drop the first 60 s of every recording
scoring	**evaluation.agg_windows**	none	mean	one prediction per recording = mean of its windows (Eq. (3.7))
optimizer	**optimizer.lr**	0.0005	0.0001	peak lr 1e-4 instead of 5e-4
schedule	**optimizer.warmup_epochs**	5	2	2 warmup epochs instead of 5
	trainer.epochs	50	15	15 epochs instead of 50

Loss functions. The loss is what the optimizer minimizes on every training batch. LaBraM as published fine-tunes TUAB as a binary normal/abnormal classifier. The head emits one logit s per window with label $c \in \{0, 1\}$, and the loss is the binary cross-entropy (`nn.BCEWithLogitsLoss`, `labram/losses/classification.py`):

$$\ell_{\mathrm{BCE}}(s, c) = -\big[c \log \sigma(s) + (1 - c) \log(1 - \sigma(s))\big], \qquad \sigma(s) = \frac{1}{1 + e^{-s}} \qquad (2.1)$$

For brain age the head instead emits a scalar $\hat{z}$ per 10 s window. The target is the z-scored age $z = (y - \mu)/\sigma$ (Eq. (3.2), with μ and σ from the training windows). The batch loss averages a per-window loss ℓ of the residual over the B windows of the batch (`labram/losses/regression.py::build_regression_criterion`):

$$\mathcal{L} = \frac{1}{B}\sum_{b=1}^{B} \ell(r_b), \qquad r_b = \hat{z}_b - z_b \tag{2.2}$$

Per-window loss. The default for D and every later run unless stated otherwise is the **Huber** loss with $\delta = 1$. It is quadratic for small residuals and linear for large ones, so a few recordings with a wrong or extreme age do not dominate the gradient. DX2 uses $\delta = 2$:

$$\ell_\delta(r) = \begin{cases} \frac{1}{2}r^2 & |r| \le \delta \\ \delta\left(|r| - \frac{1}{2}\delta\right) & \text{otherwise} \end{cases} \tag{2.3}$$

D2 uses the **L1** loss, the absolute error, which directly matches the MAE of Eq. (0.1):

$$\ell_1(r) = |r| \tag{2.4}$$

Mixup (M1 and every phase-2 run). Each window b of a batch is blended with a partner window $\pi(b)$ drawn from the same batch, usually from another recording with a different age. π permutes the **samples of the batch** (`torch.randperm` over the batch dimension), not the channels. The two windows are averaged point by point, channel with the same channel and time sample with the same time sample, and their targets with the same weight. For example, $\lambda = 0.7$ blends a 30-year-old's window with a 70-year-old's into target $0.7 \cdot 30 + 0.3 \cdot 70 = 42$. One $\lambda \sim \mathrm{Beta}(0.4, 0.4)$ is drawn per batch, usually near 0 or 1, so one window dominates. The loss of Eq. (2.2) is computed on the blended targets (`labram/train/mixup.py::mixup_batch`). A channel missing from either partner (T1/T2 in the `channel_drop` runs) is zeroed rather than blended:

$$\bar{x}_b = \lambda x_b + (1-\lambda)\, x_{\pi(b)}, \qquad \bar{z}_b = \lambda z_b + (1-\lambda)\, z_{\pi(b)} \tag{2.5}$$

Plain mixup draws the partner $\pi(b)$ uniformly at random. **C-Mixup** (M1c) prefers partners of similar age, so the blended target stays plausible. With $\sigma_m = 5$ years:

$$P\big(\pi(b) = j\big) \propto \exp\Big(-\frac{(y_b - y_j)^2}{2\sigma_m^2}\Big), \quad j \ne b \tag{2.6}$$

Soft labels (M4). The head emits a softmax p_k over age bins with centres c_k = 1, 2, ..., 89 years. The target is a Gaussian q_k around the true age with $\sigma_s = 2.5$ years, the loss is the KL divergence, and the prediction is the expected age (`SoftLabelRegressionLoss`). Centres and σ_s are z-scored like z:

$$\begin{aligned} q_k &= \frac{\exp\big(-(c_k - z)^2/2\sigma_s^2\big)}{\sum_{k'} \exp\big(-(c_{k'} - z)^2/2\sigma_s^2\big)}, \qquad \ell_{\mathrm{KL}} = \sum_k q_k \log\frac{q_k}{p_k}, \\ \hat{z} &= \sum_k p_k\, c_k \end{aligned} \tag{2.7}$$

Age-balanced loss (B1, B2). Rare ages (children, the very old) are up-weighted. ρ_k is the density of training windows in 1-year age bin k, smoothed by a Gaussian with $\sigma = 2$ years.

The weight is $\rho^{-1/2}$ (B1) or ρ^{-1} (B2), capped at 10× the smallest weight and normalized to mean 1 over the training windows (`age_balance_weights` , `BalancedRegressionLoss`):

$$\mathcal{L} = \frac{1}{B}\sum_{b} w_{k(y_b)}\,\ell_\delta(r_b), \qquad w_k \propto \rho_k^{-1/2} \text{ or } \rho_k^{-1} \tag{2.8}$$

Codebook regularization: the spectral reconstruction and quantization terms (DX1, DX4, DX4b only). LaBraM's tokenizer (VQNSP) was pre-trained to encode each 1 s patch into a codeword and to decode from it the patch's Fourier amplitude and phase. These runs reuse its quantizer and decoder, frozen, on top of the fine-tuned encoder. Extra loss terms then penalize the encoder if its patch tokens drift away from what the frozen decoder can read back (`model.codebook_reg` , `labram/losses/codebook_regularized.py`). The age head still reads the mean patch token, as in D.

For each patch token n (one channel × one 1 s patch of 200 samples):

- $\tilde{A}_{n,k}$: the std-normalized FFT amplitude of the patch at frequency bin k (bin k = k Hz);
- $\hat{A}_{n,k}$: the decoder's reconstruction of it;
- h_n: the encoder token;
- e_n: its nearest codeword; sg is stop-gradient.

The terms are averaged over all tokens (and bins or feature dimensions) of the batch:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_\delta + w_A\,\mathcal{L}_{\text{amp}} + w_P\,\mathcal{L}_{\text{phase}} + w_Q\,\mathcal{L}_{\text{q}}, \\ \mathcal{L}_{\text{amp}} &= \text{mean}_{n,\,k\le K}\,\big(\hat{A}_{n,k} - \tilde{A}_{n,k}\big)^2, \\ \mathcal{L}_{\text{q}} &= \beta_q\ \text{mean}_{n}\ \big(\text{sg}(e_n) - h_n\big)^2, \qquad \beta_q = 1 \end{aligned} \tag{2.9}$$

$\mathcal{L}_{\text{phase}}$ is the same MSE on the std-normalized FFT phase. K is the number of frequency bins kept: `loss.freq_fraction` × 200. The weights each run used, from its ClearML configuration (`model.codebook_reg.*_weight`):

- **DX1:** $w_A = 1$, $w_P = 0$, $w_Q = 0$, so amplitude reconstruction only. `freq_fraction` = 0.5 keeps $K = 100$ bins, 0–99 Hz. That is the whole non-redundant half of the spectrum: for a real signal, the upper 100 bins mirror the lower ones.
- **DX4:** $w_A = 0$, $w_P = 0$, $w_Q = 1$, so the quantization (commitment) term only. No reconstruction.
- **DX4b:** as DX4 with $w_Q = 50$.
- **Phase** (w_P) is 0 in all three. The phase reconstruction loss was never applied in any fine-tune of this study.
- **Every other run** has `model.codebook_reg.enabled=false` . The model then has no quantizer or decoder, and all three terms are absent. The `loss.amplitude_weight` / `phase_weight` / `embedding_weight` fields that read 1.0 in every run's ClearML configuration belong to the VQNSP tokenizer's own training and are never read by a fine-tune.

None of the three helped. Compared with D, the test MAE changes by +0.24 (DX1), +0.14 (DX4) and +0.36 (DX4b), all worse but all within seed noise, τ = 0.38 (Eq. (5.2)). Table 5.2 tests DX1 and DX4 in rows P17 and P18.

Multi-task (MT1). A second output predicts the normal/abnormal label c with the BCE of Eq. (2.1), weighted 0.5 (`MultiTaskAgeCriterion`). Selection still uses the age MAE:

$$\mathcal{L} = \mathcal{L}_\delta + 0.5\,\frac{1}{B}\sum_b \ell_{\mathrm{BCE}}(s_b, c_b) \tag{2.10}$$

Channel-dropout consistency (Exp4aug, Exp4aug15, cvaug). With probability 0.2 per batch, 1–2 of the T1/T2 channels are dropped from every sample. The age loss is computed on that dropped pass. The feature f the head reads should not change: it is pulled towards the feature f^{full} of a frozen all-channels pass, with $\lambda = 0.1$, over the samples A that lost a channel and the feature dimensions $d = 1..D$ (`labram/train/channel_drop.py`):

$$\mathcal{L} = \mathcal{L}_\delta^{\mathrm{drop}} + \lambda\,\frac{1}{|A|D}\sum_{b\in A}\sum_{d=1}^{D}\left(f_{b,d}^{\mathrm{full}} - f_{b,d}^{\mathrm{drop}}\right)^2 \tag{2.11}$$

Here $\mathcal{L}_\delta$ is Eq. (2.2) with the Huber loss of Eq. (2.3). Changes that do not touch the loss (EMA evaluation, layer decay, dropout, heads, input conditioning) keep Huber $\delta = 1$, with mixup wherever the run's `loss` column says so.

Table 2.2. Phase 1: TUAB-only runs

The experiment registry's master table (`docs/age_overfitting_experiments.md` , Table 12.1).

Columns:

- `ClearML` : the run's ClearML experiment, linked by its name.
- `family` and `change` : the question the run belongs to, and what it changes relative to D unless it names another run. Abbreviations:
  - CAR common average reference; ep epochs; lr learning rate;
  - ld layer decay; dp drop path; drop dropout; wd weight decay; clip gradient clipping at norm 1;
  - EMA evaluation with an exponential moving average of the weights;
  - LoRA r16 low-rank adapters of rank 16 instead of full fine-tuning;
  - CLS the class token; MLP a two-layer head.
- `loss` : the run's training loss, read from its ClearML configuration (Eqs. (2.2)–(2.11)).
- `val` / `test` : MAE (Eq. (0.1), years) at the epoch with the lowest validation MAE. Phase 1 is **TUAB only**, in training and evaluation:
  - `val` : TUAB's validation split, 556 recordings (291 normal, 265 abnormal);
  - `test` : TUAB's official evaluation set, 274 recordings.

- `N` / `A`: the same TUAB test MAE restricted to its 149 normal and 125 abnormal recordings.
- `β`: the bias slope of Eq. (0.5) on the 274 TUAB test recordings. Every model regresses towards the mean age. The fine-tuned runs sit between −0.46 and −0.33, and the from-scratch runs S1–S3 at −0.49 to −0.52. The collapsed runs (C2, C2-head, C2-last, DX3, F2; test MAE above 13) reach −0.8 to −1.0, meaning they predict almost a constant age.
- `e*/E`: the selected epoch $e^\star$ and the number of epochs E.

Same data split in every run. Every run records the recordings of its train, val and test splits (the `data_split` artifact of its ClearML experiment), saved here as SHA-1 digests of the sorted recording lists (`docs/data/brain_age_run_splits.csv`, `rs.split_fingerprint`, regenerated by `rs.fetch_run_splits`). The check below the table compares them: all phase-1 runs train, validate and test on identical recordings, with no patient in two splits. Only the number of windows differs, where the run changes the windows: A scores whole recordings, and C uses 30 s windows. Phase 2's `Control` uses the same split.

Why some `N` / `A` are missing ("–"). The per-cohort test metrics were added to the trainer on 2026-10-07 06:47 UTC (commit `62b159d`). 16 of the 21 runs without them started before that. The other five (E2, DX3, E1, E1c, E2c) are SageMaker jobs started 07:06–08:30 the same day whose job code did not include the change yet. All 21 logged only the overall val/test MAE. They saved no per-recording predictions, so the split by status cannot be recomputed from their logs; it needs a new inference pass of their checkpoints. Their `test` column is on the same 274 recordings as everyone else's.

The pre-study baseline (LaBraM fine-tuned with the original recipe adapted to regression) scored val 8.31 / test 9.04 (`docs/age_training_scenario_D.md`, Section 1).

run	ClearML	family	change	loss	val	test	N	A	β	e*/E
A	age_A_ep15_lr1e-4_fullrec_20261005_043322_643	Evaluation protocol	B with whole-recording val/test	Huber δ 1 (2.3)	8.29	9.03	–	–	-0.46	2/12
B	age_B_eval5min_20261005_064356_435	Evaluation protocol	CAR off, 15 ep, lr 1e-4, 5-min eval (pre-D baseline)	Huber δ 1 (2.3)	8.40	8.91	–	–	-0.43	2/6
D	age_D_eval5min_car_20261005_013250_973	Baseline	B + CAR (finetune_tuab_age.json)	Huber δ 1 (2.3)	7.77	8.30	–	–	-0.40	1/15
D-npy	age_D_eval5min_car_npy_file_20261006_185443_758	Baseline (repro)	D on the npy format	Huber δ 1 (2.3)	7.77	8.30	–	–	-0.40	1/15
D-cohorts	age_D_npy_cohorts_20261007_064710_242	Baseline (repro)	D, local, cohort metrics	Huber δ 1 (2.3)	7.77	8.31	6.57	10.38	-0.40	1/15
E2	age_E2_npy_ep4_warm1_20261007_070635_244	Schedule	D, 4 ep / 1 warmup	Huber δ 1 (2.3)	7.68	8.25	–	–	-0.37	3/4
L1	age_L1_ep12_mixup_ema_20261009_095540_331	Schedule	M12 on 12 ep / 2 warmup	Huber δ 1 + mixup (2.3, 2.5)	7.51	8.47	6.65	10.63	-0.42	3/12
C2	age_C2_eval5min_regularized_20261005_012652_087	Classic regularizers	B + ld 0.5, dp 0.2, drop 0.1, wd 0.1, clip 1	Huber δ 1 (2.3)	12.47	13.52	–	–	-0.81	14/15
DX3	age_DX3_npy_regularized_20261007_070603_726	Classic regularizers	D + C2 bundle	Huber δ 1 (2.3)	12.83	14.90	–	–	-1.00	0/15
F1	age_F1_short_ld05_20261008_215552_344	Classic regularizers	E2 + layer decay 0.5	Huber δ 1 (2.3)	7.74	8.36	6.42	10.68	-0.36	3/4
F2	age_F2_short_drop01_20261008_230105_724	Classic regularizers	E2 + dropout 0.1	Huber δ 1 (2.3)	13.25	15.17	13.37	17.32	-0.97	0/4
F3	age_F3_short_dp02_20261008_232500_075	Classic regularizers	E2 + drop path 0.2	Huber δ 1 (2.3)	7.70	8.41	6.31	10.91	-0.35	3/4
F4	age_F4_short_wd01_20261008_232504_307	Classic regularizers	E2 + weight decay 0.1	Huber δ 1 (2.3)	7.66	8.24	6.49	10.32	-0.37	3/4
G1	age_G1_short_init1_20261008_230259_560	Head init	E2 + init_scale 1.0	Huber δ 1 (2.3)	7.76	8.38	6.64	10.45	-0.37	3/4
H1	age_H1_short_ld075_20261009_003609_895	Layer decay	E2 + layer decay 0.75	Huber δ 1 (2.3)	7.71	8.67	6.64	11.09	-0.40	1/4
H2	age_H2_short_ld085_20261009_003617_662	Layer decay	E2 + layer decay 0.85	Huber δ 1 (2.3)	7.86	8.54	6.74	10.69	-0.42	3/4
C2-head	age_C2_eval5min_regularized_headonly_20261004_220639_025	Partial fine-tuning	C2, head only	Huber δ 1 (2.3)	12.77	14.78	–	–	-0.99	5/8
C2-last	age_C2_eval5min_regularized_lastblock_head_20261005_000707_645	Partial fine-tuning	C2, last block + head	Huber δ 1 (2.3)	12.50	13.64	–	–	-0.87	6/15
M3	age_M3_short_lora16_20261008_215551_826	Partial fine-tuning	E2 + LoRA r16, lr 1e-3, ld 1.0	Huber δ 1 (2.3)	7.68	9.24	7.41	11.42	-0.44	3/4
M1	age_M1_short_mixup04_20261007_231716_154	Mixup / EMA	E2 + mixup 0.4	Huber δ 1 + mixup (2.3, 2.5)	7.46	8.45	6.60	10.65	-0.41	3/4
M1c	age_M1c_short_cmixup04_s5_20261008_001050_123	Mixup / EMA	E2 + C-Mixup 0.4, σ 5 y	Huber δ 1 + C-Mixup σ 5 y (2.3, 2.5, 2.6)	7.54	8.23	6.50	10.29	-0.39	3/4
M2	age_M2_short_ema_20261008_212509_991	Mixup / EMA	E2 + EMA eval (0.9995)	Huber δ 1 (2.3)	7.58	8.17	6.35	10.35	-0.36	2/4
M12	age_M12_short_mixup_ema_20261008_204214_769	Mixup / EMA	E2 + mixup 0.4 + EMA eval	Huber δ 1 + mixup (2.3, 2.5)	7.48	8.45	6.57	10.69	-0.42	3/4
D2	age_D2_eval5min_l1_20261005_000636_646	Loss	B + L1 loss	L1 (2.4)	8.36	9.32	–	–	-0.46	6/15
DX2	age_DX2_huber_delta2_20261005_185121_984	Loss	D + Huber δ 2	Huber δ 2 (2.3)	7.79	8.41	–	–	-0.38	3/15
M4	age_M4_short_softlabel_20261008_215551_240	Loss	E2 + soft labels σ 2.5 y	soft-label KL, σ 2.5 y (2.7)	7.67	8.59	6.92	10.57	-0.40	3/4
B1	age_B1_short_balance_sqrtinv_20261009_124653_371	Loss	E2 + age-balanced, sqrt-inverse	Huber δ 1, age weights ρ^-1/2 (2.8)	7.83	8.48	6.66	10.63	-0.37	2/4
B2	age_B2_short_balance_inv_20261009_134321_186	Loss	E2 + age-balanced, inverse	Huber δ 1, age weights ρ^-1 (2.8)	7.96	8.57	6.70	10.80	-0.33	1/4
DX1	age_DX1_amp_freq05_20261006_022243_393	Codebook regularization	D + amplitude reconstruction (½ spectrum)	Huber δ 1 + amplitude reconstruction, w 1 (2.9)	7.72	8.54	–	–	-0.39	4/15
DX4	age_DX4_quant_20261005_160655_990	Codebook regularization	D + quantization term, weight 1	Huber δ 1 + quantization, w 1 (2.9)	7.67	8.44	–	–	-0.39	3/15
DX4b	age_DX4b_quant_w50_20261005_215817_973	Codebook regularization	D + quantization term, weight 50	Huber δ 1 + quantization, w 50 (2.9)	7.78	8.66	–	–	-0.37	4/15
C	age_C_eval5min_win30s_20261004_235942_658	Augmentation / input	B + 30-s windows	Huber δ 1 (2.3)	8.24	9.35	–	–	-0.42	8/15
D-clip1	age_D_npy_clip1_20261007_003433_521	Augmentation / input	D + clip 1.0	Huber δ 1 (2.3)	7.70	8.36	–	–	-0.41	1/15
E1	age_E1_npy_random_crop_20261007_070629_845	Augmentation / input	D + random crop	Huber δ 1 (2.3)	7.82	8.56	–	–	-0.40	3/15
E1c	age_E1c_npy_random_crop_clip1_20261007_070930_530	Augmentation / input	D + random crop + clip 1.0	Huber δ 1 (2.3)	7.72	8.13	–	–	-0.38	1/15
E2c	age_E2c_npy_ep4_warm1_clip1_20261007_083022_212	Augmentation / input	E2 + clip 1.0	Huber δ 1 (2.3)	7.63	8.24	–	–	-0.38	3/4
D-cls	age_D_npy_cls_head_20261007_003458_637	Head	D, CLS token instead of mean pooling	Huber δ 1 (2.3)	7.90	8.24	–	–	-0.38	2/15
D-clsmean	age_D_npy_cls_mean_head_20261007_003450_736	Head	D, [mean patch; CLS] concat	Huber δ 1 (2.3)	7.80	8.35	–	–	-0.41	1/15

run	ClearML	family	change	loss	val	test	N	A	β	e*/E
N2L	age_N2L_ep12_mixup04_clip1_attnhead_20261009_232420_478	Head	D + mixup 0.4 + clip 1 + attention head, 12 ep / 4 warmup	Huber δ 1 + mixup (2.3, 2.5)	7.63	8.49	6.68	10.64	-0.43	4/12
MT1	age_MT1_short_mixup04_multitask_20261010_234501_492	Head (multi-task)	M1 + age/abnormal multi-task head (seed 0)	Huber δ 1 + mixup + 0.5 BCE abnormal (2.3, 2.5, 2.10)	7.54	8.38	6.73	10.35	-0.40	3/4
MT1-seed1	age_MT1_short_mixup04_multitask_seed1_local_20261011_003814_186	Head (multi-task)	MT1, seed 1	Huber δ 1 + mixup + 0.5 BCE abnormal (2.3, 2.5, 2.10)	7.62	8.54	6.75	10.69	-0.41	3/4
S1	age_S1_scratch_ep15_20261010_084032_716	From scratch	D, no pretrained weights, ld 1.0	Huber δ 1 (2.3)	8.79	9.42	7.75	11.42	-0.49	14/15
S2	age_S2_scratch_d6_mlp_ep15_20261010_092116_377	From scratch	scratch, 6 blocks + MLP head, tuned (seed 0)	Huber δ 1 + mixup (2.3, 2.5)	8.53	9.30	7.48	11.48	-0.50	12/15
S2-seed1	age_S2_scratch_d6_mlp_ep15_seed1_local_20261010_142248_698	From scratch	S2, seed 1	Huber δ 1 + mixup (2.3, 2.5)	8.58	9.44	7.84	11.35	-0.50	14/15
S3	age_S3_scratch_d6_mlp_aug_ep15_local_20261010_215949_465	From scratch	S2 + label-preserving augmentation + ±800 µV clip	Huber δ 1 + mixup (2.3, 2.5)	8.93	9.59	7.91	11.59	-0.52	9/15

`Split check: all 45 phase-1 runs use the same recordings: train 2,145 (1,650 patients), val 556 (412), test 274 (251). Window counts differ only for A, C. Same split outside phase 1: Control.`

Table 2.3. Phase 2: TUH cohort runs

One row per run of `docs/data/tuh_cohort_eval_v1.csv`, from `labram/eval/research_summary.py::PHASE2`:

- `ClearML`: the run's ClearML experiment, linked by its name;
- `cohort` (Table 1.1), `epochs` / `warmup`;
- `cond.`: ✓ if the head also receives the recording's status and pathology vector (Eq. (3.6));
- `parent` and `change`: the run it is compared with in Section 5 and what differs;
- `loss`: the training loss (Eqs. (2.2)–(2.11)). Every phase-2 run is M1, Huber + mixup; the T1/T2 dropout runs add the consistency term.

Metrics:

- `test all` / `normal` / `abnormal`: MAE (Eq. (0.1)) on the **TUAB test set only**, all 274 recordings and its 149 normal / 125 abnormal ones. The phase-2 runs train on TUEG too, but none of them is tested on TUEG recordings: TUAB test is the one set every run shares (Section 0).
- `R² normal`: Eq. (0.2) on the normal TUAB test recordings;
- `TUEG val v2`: MAE on the TUEG part of the shared v2 validation split (2,325 recordings no v2/v3 model trained on), available only for runs scored on it. It is the only TUEG number in the table. Each run's own validation split is not shown, because it differs from cohort to cohort.
- No `β`: the cohort evaluation (`docs/data/tuh_cohort_eval_v1.csv`) stores MAE, RMSE and R² but not the bias slope, so β is reported for phase 1 only.

Same split or not. The check below the table groups the runs by their recorded train/val/test recordings, as for Table 2.2:

- Runs on the same cohort share one split. Exp1f's `normal_both_v1_meta` holds the same recordings as `normal_both_v1`, and `all_both_v3_compact` the same as `all_both_v3`.

- **Exp3q is the exception** within `all_both_v1`. Its signal-quality filter drops windows beyond ±800 µV and windows with a flat channel, and recordings left without windows drop out. That leaves 19,815 instead of 20,368 train recordings and 2,836 instead of 2,905 val recordings, and its test loses 10 windows but keeps all 274 recordings.
- Splits differ across cohorts by design (Table 1.1): `v1`, `v2` and `v3` assign TUEG patients differently, so their val MAEs do not compare.
- Every run with a test split tests on the same 274 TUAB recordings. Exp2 (`normal_TUEG_v1`) has no test split.

run	ClearML	cohort	epochs	cond.	parent	change	loss	test all	nor-mal	ab-nor-mal	R^2 nor-mal	TUEG val v2
Control	tuh_control_tuab_only_M1_interp_20261012_152904_458	TUAB_v3.0.1_interp50-60	4 / 1			M1 on TUAB only (the TUAB baseline of phase 2)	Huber δ 1 + mixup (2.3, 2.5)	8.23	6.42	10.39	0.75	9.29
Exp1	tuh_exp1_normal_both_M1_20261012_133302_883	normal_both_v1	4 / 1		Control	train on normal recordings only: TUAB normal + TUEG normal	Huber δ 1 + mixup (2.3, 2.5)	8.03	5.99	10.47	0.78	8.68
Exp2	tuh_exp2_normal_tueg_M1_20261012_134712_671	normal_TUEG_v1	4 / 1		Exp1	TUEG normal only (no TUAB in training)	Huber δ 1 + mixup (2.3, 2.5)	8.00	6.06	10.31	0.78	8.75
Exp1b	tuh_exp1b_normal_both_M1_ep10_20261012_175842_602	normal_both_v1	10 / 3		Exp1	10 epochs / 3 warmup instead of 4 / 1	Huber δ 1 + mixup (2.3, 2.5)	8.11	6.09	10.52	0.77	8.66
Exp1t	tuh_exp1t_normal_both_M1_ep4_feat-tokens_20261013_091030_953	normal_both_v1	4 / 1		Exp1	classical EEG features as extra transformer tokens	Huber δ 1 + mixup (2.3, 2.5)	8.80	7.17	10.73	0.70	–
Exp1f	tuh_exp1f_normal_both_M1_ep4_feat-only_20261012_235557_718	normal_both_v1_meta	4 / 1		Exp1	classical EEG features only, MLP head, no EEG encoder	Huber δ 1 + mixup (2.3, 2.5)	8.13	6.21	10.41	0.77	–
Exp3	tuh_exp3_all_both_M1_20261012_140234_136	all_both_v1	4 / 1		Control	add every TUEG recording (normal, abnormal, unlabelled): ~10x training data	Huber δ 1 + mixup (2.3, 2.5)	7.26	5.57	9.28	0.80	–
Exp3b	tuh_exp3b_all_both_M1_ep10_20261012_205133_238	all_both_v1	10 / 3		Exp3	10 epochs / 3 warmup instead of 4 / 1	Huber δ 1 + mixup (2.3, 2.5)	7.70	6.17	9.51	0.75	–
Exp3q	tuh_exp3q_all_both_M1_ep4_qc800-0.5_20261013_110534_570	all_both_v1	4 / 1		Exp3	drop windows beyond ±800 µV or with a flat channel	Huber δ 1 + mixup (2.3, 2.5)	7.34	5.69	9.31	0.79	–
Exp3c	tuh_exp3c_all_both_M1_ep4_clip800_20261013_103559_425	all_both_v1	4 / 1		Exp3	clip the input to ±800 µV inside the model	Huber δ 1 + mixup (2.3, 2.5)	7.29	5.66	9.22	0.79	–
Exp3v2	tuh_exp3_all_both_v2_M1_ep4_20261013_202310_767	all_both_v2	4 / 1		Exp3	cohort v2: one TUEG patient split shared by every cohort (data bookkeeping only)	Huber δ 1 + mixup (2.3, 2.5)	7.42	5.46	9.75	0.80	8.00
Exp3cond	tuh_exp3cond_all_both_v2_M1_ep4_cond-status-pathology_20261013_221741_619	all_both_v2	4 / 1	✓	Exp3v2	status + pathology vectors concatenated to the pooled feature	Huber δ 1 + mixup (2.3, 2.5)	7.28	5.39	9.53	0.81	7.90
Exp3clscond	tuh_exp3clscond_all_both_v2_M1_ep4_cls-cond-status-pathology_20261014_094752_639	all_both_v2	4 / 1	✓	Exp3cond	head on the class token instead of the mean patch token	Huber δ 1 + mixup (2.3, 2.5)	7.25	5.36	9.51	0.81	7.98
Exp4drop	tuh_exp4_dropT1T2_all_both_v3_cond_M1_ep4_20261014_093428_304	all_both_v3	4 / 1	✓	Exp3cond	cohort v3 (+ recordings missing only T1/T2) on 21 channels (T1/T2 dropped everywhere)	Huber δ 1 + mixup (2.3, 2.5)	7.28	5.26	9.68	0.82	7.76
Exp4base	tuh_exp4base_all_both_v3_M1_ep6_base-fast_20261014_182259_096	all_both_v3	6 / 2		Exp3v2	cohort v3, 23 channels with missing T1/T2 left out per sample; speed settings; 6 / 2 epochs	Huber δ 1 + mixup (2.3, 2.5)	7.36	5.62	9.44	0.81	7.86
Exp4basec	tuh_exp4basec_all_both_v3_compact_M1_ep6_base-fast-compact_20261015_222728_648	all_both_v3_compact	6 / 2		Exp4base	compact copy of the cohort, same recorded split (reproduction check)	Huber δ 1 + mixup (2.3, 2.5)	7.42	5.52	9.68	0.81	7.85
Exp4ctrl	tuh_exp4ctrl_all_both_v3_M1_ep10_cls-cond-ctrl-fast_20261015_034811_426	all_both_v3	10 / 3	✓	Exp4base	class token + condition vectors, 10 / 3 epochs with early stopping	Huber δ 1 + mixup (2.3, 2.5)	7.30	5.35	9.63	0.81	7.92
Exp4aug	tuh_exp4_augT1T2_v3_clscond_ep6_lam0.1_b128_20261014_180323_818	all_both_v3	6 / 2	✓	Exp4ctrl	T1/T2 batch dropout (p 0.2) + consistency loss (λ 0.1); 6 / 2 epochs	Huber δ 1 + mixup + 0.1 consistency (2.3, 2.5, 2.11)	7.29	5.45	9.49	0.82	7.85
Exp4aug15	tuh_exp4_augT1T2_v3_clscond_ep15_lam0.1_b128_20261015_030018_655	all_both_v3	15 / 5	✓	Exp4aug	15 / 5 epochs with early stopping (stopped at 11)	Huber δ 1 + mixup + 0.1 consistency (2.3, 2.5, 2.11)	7.17	5.38	9.29	0.81	7.91

```
same split: Exp1, Exp1b, Exp1t, Exp1f  (cohort normal_both_v1, normal_both_v1_meta)
same split: Exp3v2, Exp3cond, Exp3clscond  (cohort all_both_v2)
same split: Control  (cohort TUAB_v3.0.1_interp50-60)
same split: Exp4drop, Exp4base, Exp4basec, Exp4ctrl, Exp4aug, Exp4aug15  (cohort all_bo
th_v3, all_both_v3_compact)
same split: Exp3q  (cohort all_both_v1)
same split: Exp3, Exp3b, Exp3c  (cohort all_both_v1)
same split: Exp2  (cohort normal_TUEG_v1)
TUAB test recordings identical in every phase-2 run with a test split: True
```

Table 2.4. Phase 3: five-fold cross-validation

- **Folds:** the pool is `all_both_v3_cvcompact` train + val (29,035 recordings), split into 5 patient-disjoint folds. For fold k, test = fold k, val = fold $k+1 \bmod 5$, and the other three train. The TUAB test set is in no fold, so each of the 5 fold models also scores it.
- **`ClearML`:** the ClearML project holding the run's five fold experiments, linked by its name.
- **`loss`:** the training loss (Eqs. (2.2)–(2.11)).
- **Per-fold columns:** mean ± sample sd over the 5 fold models of the MAE (Eq. (0.1)) on:
  - `CV val`: the fold's own val;
  - `CV test`: the fold's own test;
  - `TUAB test`, with `normal` and `abnormal`.
- **`ens.`:** the TUAB test MAE of the median of the 5 models' predictions (Eq. (3.8)). It needs the per-recording predictions.

Source: `docs/data/cv5_eval_v2.csv`.

run	ClearML	cond.	loss	CV val	CV test	TUAB test	nor-mal	abnor-mal	ens. TUAB test	ens. normal	ens. abnor-mal
cvbase	tuh_cvbase_all_both_v3_cvcompact_M1_ep5_cv5-base_cv5		Huber δ 1 + mixup (2.3, 2.5)	8.16 ± 0.11	8.18 ± 0.15	7.49 ± 0.09	5.81 ± 0.10	9.50 ± 0.11	7.44	5.75	9.46
cvctrl	tuh_cvctrl_all_both_v3_cvcompact_M1_ep10_cv5-ctrl_cv5	✓	Huber δ 1 + mixup (2.3, 2.5)	8.09 ± 0.12	8.08 ± 0.13	7.28 ± 0.14	5.53 ± 0.08	9.37 ± 0.25	7.10	5.31	9.23
cvaug	tuh_cvaug_all_both_v3_cvcompact_M1_ep10_cv5-augT1T2_cv5	✓	Huber δ 1 + mixup + 0.1 consistency (2.3, 2.5, 2.11)	8.06 ± 0.13	8.06 ± 0.15	7.33 ± 0.17	5.55 ± 0.17	9.44 ± 0.18	7.23	5.43	9.37

Figure 2.1. TUAB test MAE of every run

Every run's TUAB test MAE (Eq. (0.1), years), in the order the runs were made, left to right. Dotted vertical lines separate phase 1 (Table 2.2), phase 2 (Table 2.3) and the CV runs (Table 2.4).

- (a) All 274 test recordings.
- (b) The 149 normal test recordings.
- (c) The 125 abnormal test recordings.

The 21 phase-1 runs without cohort metrics (Table 2.2) have no point in (b) or (c).

Legend.

- **Colour = phase:** blue phase 1 (TUAB only), orange phase 2 (TUH cohorts), green phase 3 (cross-validation).
- **Fill = what the head sees:**
  - filled marker: an EEG-only model;
  - hollow marker: the head also receives the recording's status and pathology vector (Eq. (3.6)), so the model is not EEG-only and its MAE is not comparable with published results.
- **Shape:**
  - circle: one model;
  - green circle with bars: the mean ± sample sd over the 5 CV fold models;
  - green star: the median ensemble of those 5 models (Eq. (3.8));
  - triangle: a collapsed run whose MAE is above the panel's top edge (11 years in (a) and (b), 14 in (c)). It is drawn at the edge so the axis stays readable; its true value is in Table 2.2.
- **Published results, black horizontal lines.** Both are the best published TUAB brain-age models. Their training data and splits differ from ours, so the lines are reference points, not a matched comparison.
  - Dashed: Gemein et al. 2024, a TCN ensemble trained on TUEG normal recordings only. On TUAB's evaluation set it scores 6.60 on normal recordings (b) and 12.85 on abnormal ones (c), **after** their age-bias correction; their raw normal MAE was 6.48. Our points are raw, and Table 3.5 gives our numbers after the same kind of correction. Their abnormal error comes from a model that never saw an abnormal recording; ours do.
  - Dotted: Engemann et al. 2022, the best earlier result, 7.75 on TUAB normal recordings (b) (10-fold cross-validation).
  - Panel (a) has no line: neither paper reports one MAE over all TUAB recordings.

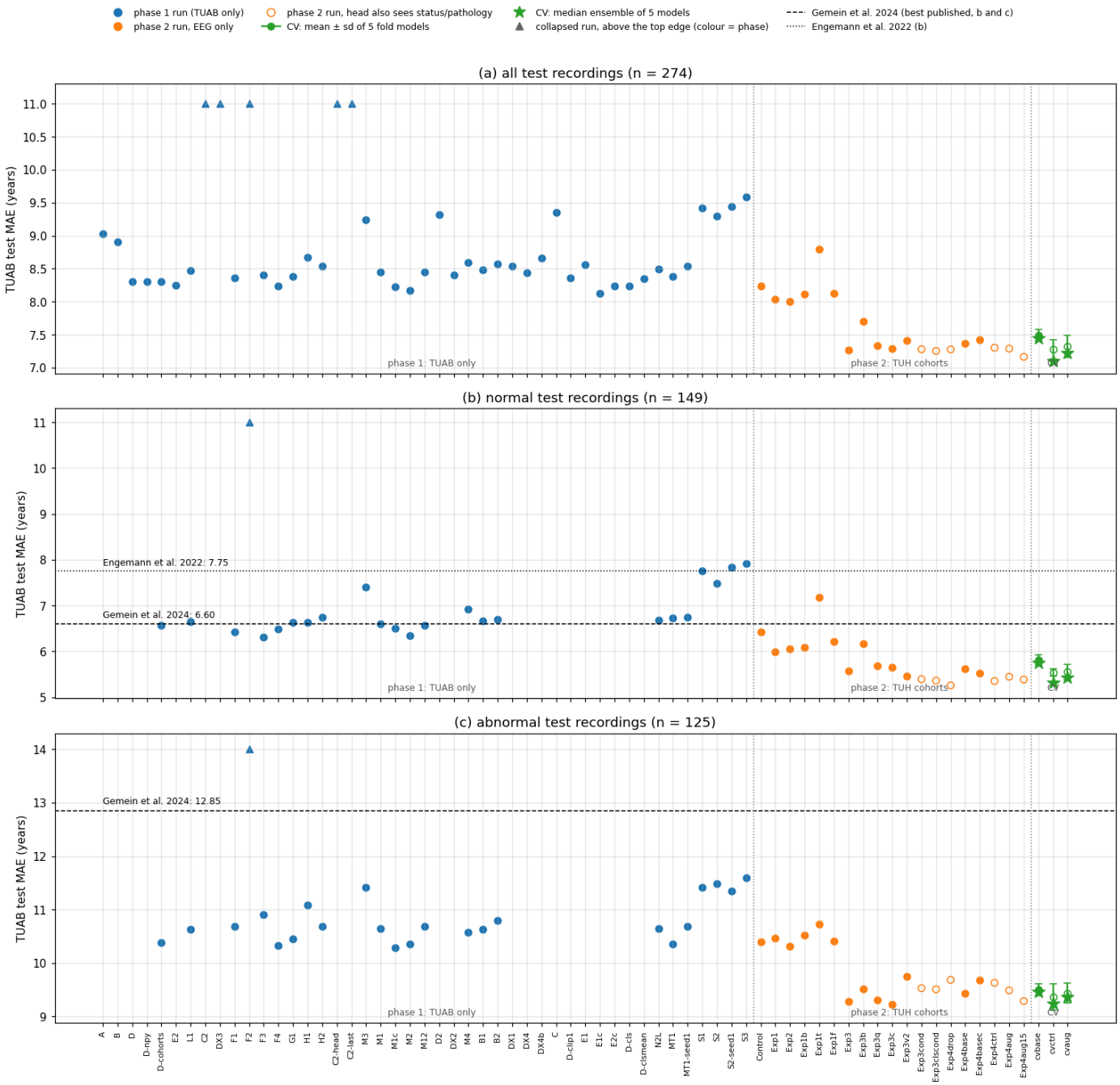


Table 2.5. The runs around the best result

The 12 runs with the lowest TUAB test MAE across all phases (Tables 2.2–2.4):

- `kind`:
  - `single split`: one run;
  - `CV mean`: mean over 5 fold models;
  - `CV ensemble`: median of 5 fold models.
- `cond.`: ✓ if the head is given the status/pathology vector.
- `ClearML`: the run's ClearML experiment (for CV: the project of its folds), linked by its name.

- `test all` / `normal` / `abnormal`: MAE (Eq. (0.1)) on the **TUAB test set only**, all 274 recordings and its 149 normal / 125 abnormal ones. The same set is used in every kind of row: for the CV rows it is the held-out TUAB test set, which no fold trains on (Table 2.4). The status is TUAB's own normal/abnormal label.

Ranked on `test all`, with the best EEG-only run in each kind marked `◆`.

run	ClearML	kind	phase	cond.	test all	nor-mal	abnor-mal
cvctrl	tuh_cvctrl_all_both_v3_cvcompact_M1_ep10_cv5-ctrl_cv5	CV ensemble	3	✓	7.10	5.31	9.23
Exp4aug15	tuh_exp4_augT1T2_v3_clscond_ep15_lam0.1_b128_20261015_030018_655	single split	2	✓	7.17	5.38	9.29
cvaug	tuh_cvaug_all_both_v3_cvcompact_M1_ep10_cv5-augT1T2_cv5	CV ensemble	3	✓	7.23	5.43	9.37
Exp3clscond	tuh_exp3clscond_all_both_v2_M1_ep4_cls-cond-status-pathology_20261014_094752_639	single split	2	✓	7.25	5.36	9.51
◆ Exp3	tuh_exp3_all_both_M1_20261012_140234_136	single split	2		7.26	5.57	9.28
Exp4drop	tuh_exp4_dropT1T2_all_both_v3_cond_M1_ep4_20261014_093428_304	single split	2	✓	7.28	5.26	9.68
Exp3cond	tuh_exp3cond_all_both_v2_M1_ep4_cond-status-pathology_20261013_221741_619	single split	2	✓	7.28	5.39	9.53
cvctrl	tuh_cvctrl_all_both_v3_cvcompact_M1_ep10_cv5-ctrl_cv5	CV mean	3	✓	7.28	5.53	9.37
Exp3c	tuh_exp3c_all_both_M1_ep4_clip800_20261013_103559_425	single split	2		7.29	5.66	9.22
Exp4aug	tuh_exp4_augT1T2_v3_clscond_ep6_lam0.1_b128_20261014_180323_818	single split	2	✓	7.29	5.45	9.49
Exp4ctrl	tuh_exp4ctrl_all_both_v3_M1_ep10_cls-cond-ctrl-fast_20261015_034811_426	single split	2	✓	7.30	5.35	9.63
cvaug	tuh_cvaug_all_both_v3_cvcompact_M1_ep10_cv5-augT1T2_cv5	CV mean	3	✓	7.33	5.55	9.44

3. The best experiment

Which run is best. Single-split runs were compared on the TUAB test set itself (Section 0). Only the CV runs have an error bar, and their CV test / TUEG out-of-fold errors were never used for selection. So "best" is decided among the CV runs:

- **Best overall:** `cvctrl` (= Exp4ctrl cross-validated). It has the lowest TUAB test MAE, as a single fold model and as a 5-model ensemble (Table 2.4).
  - `cvaug` adds T1/T2 channel-dropout augmentation and ties with it: the differences are within noise in Table 5.2 (C2).
  - The simpler `cvctrl` is therefore the recommended configuration.
- **Best EEG-only:** `cvbase`**.** Its head sees only the EEG. `cvctrl` and `cvaug` also see the recording's status (normal / abnormal / no label) and 13 diagnosis flags from the clinical sheet. A deployed brain-age model would not have these, and published TUAB results do not use them. `cvbase` is therefore the number to compare with the literature (Table 3.3).

Both are LaBraM-base fine-tuned on `all_both_v3` with the M1 recipe. They differ in three settings (Table 3.2): class-token vs mean pooling, the condition vector, and 10 vs 5 epochs.

What the model computes

Input and target. The raw EEG u (µV) of a 10 s window is scaled to $x = u/100$ and cut into 1 s patches $x_{c,a,t}$:

- channel $c = 1..C$, with $C = 23$ minus the missing T1/T2 of that recording;
- patch $a = 1..10$, sample $t = 1..200$.

The common average reference removes what all channels share (`labram/data/preprocess.py::common_average_reference`):

$$\tilde{x}_{c,a,t} = x_{c,a,t} - \frac{1}{C}\sum_{c'=1}^{C} x_{c',a,t} \tag{3.1}$$

The age y (years) is z-scored with the training windows' mean μ and standard deviation σ (`labram/data/tuh_datasets.py::TUABAgeLoader._age_target`):

$$z = \frac{y - \mu}{\sigma} \tag{3.2}$$

Mixup. Each training batch is blended with a permuted copy of itself, using one $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$ per batch with $\alpha = 0.4$ (`labram/train/mixup.py::mixup_batch`). The condition vector c travels with the target, so it is blended too:

$$\bar{x} = \lambda x + (1-\lambda)\, x_\pi, \qquad \bar{z} = \lambda z + (1-\lambda)\, z_\pi, \qquad \bar{c} = \lambda c + (1-\lambda)\, c_\pi \tag{3.3}$$

Loss. Huber loss (Eq. (2.3)) with $\delta = 1$ on the z-scored residual $r = \hat{z} - \bar{z}$ (`labram/losses/regression.py::build_regression_criterion`):

$$\ell_\delta(r) = \begin{cases} \frac{1}{2} r^2 & |r| \le \delta \\ \delta\,(|r| - \frac{1}{2}\delta) & \text{otherwise} \end{cases} \tag{3.4}$$

Encoder and head. f_θ is the 12-block LaBraM-base transformer, initialized from the released `labram-base.pth`. Its output is $h \in \mathbb{R}^{200}$:

- the final LayerNorm's class token (`cvctrl`, `cvaug`);
- the mean over patch tokens (`cvbase`).

$$h = f_\theta(\tilde{x}) \tag{3.5}$$

The linear head reads h, concatenated with a 32-d projection of the 17-d condition vector $c \in \{0,1\}^{17}$ (3 one-hot status + 14 multi-hot pathology groups, `conditions_v1.npz`) when conditions are on (`labram/models/neural_transformer.py`, `model.session_features=head`):

$$\hat{z} = w^\top \begin{bmatrix} h \\ W_c\, c \end{bmatrix} + b, \qquad W_c \in \mathbb{R}^{32\times 17} \tag{3.6}$$

Optimizer and schedule. AdamW with weight decay 0.05. Layer-wise lr decay $\rho = 0.65$ gives block l of $L = 12$ the rate $\mathrm{lr} \cdot \rho^{L+1-l}$ (`labram/optim_factory.py::LayerDecayValueAssigner`). The rate warms up linearly, then follows a cosine down to 10^{-6} (`labram/utils/training.py`).

Scoring. A recording's prediction is the de-normalized mean over its evaluation windows W_i (the first 5 minutes after the trim, 30 windows; `evaluation.agg_windows=mean`):

$$\hat{y}_i = \mu + \sigma \frac{1}{|W_i|} \sum_{w \in W_i} \hat{z}_w \tag{3.7}$$

The fold ensemble takes the median over the $K = 5$ fold models (`labram/eval/cv_compare.py::ensemble`):

$$\hat{y}_i^{\text{ens}} = \text{median}_{k=1..K}\ \hat{y}_{i,k} \tag{3.8}$$

Table 3.1. The best configuration (`cvctrl`), step by step

One row per stage of the pipeline:

- `setting`: the configuration field;
- `cvctrl`: its value, read from the configuration rebuilt by `rs.resolve_config` (`finetune_tuab_age.json` + the run's overrides, exactly as the trainer builds it);
- `cvbase`: the EEG-only variant's value, where it differs;
- `equation`: where the setting appears above.

stage	what	setting	cvctrl	cvbase	equation
data	cohort	data.data_path	all_both_v3_cvcompact		
	5-fold CV, grouped by	cross_validation.split_by	subject		
	CV pool	cross_validation.pool	train_val		
windows	window length (s)	data.window_sec	10		
	trim start / end (s)	data.trim_start_sec / data.trim_end_sec	60 / 60		
	train minutes per recording per epoch	data.train_minutes_per_epoch	5		
	eval minutes per recording	data.eval_minutes	5		(3.7)
channels	channels left out where missing	channel_drop.channels	["T1", "T2"]		
	T1/T2 dropout probability (augmentation)	channel_drop.prob	0		
input	common average reference	labram_plus.common_average_reference	True		(3.1)
	per-patch z-scoring	labram_plus.z_score_patches	False		
encoder	model	model.model	labram_base_patch200_200		(3.5)
	initial weights	finetune_checkpoint.finetune	labram-base.pth		
	drop path	model.drop_path	0.1		
	fused attention (speed only)	model.fused_attention	True		
head	mean pooling (False = class token)	model.use_mean_pooling	False	True	(3.5)
	condition vector	model.session_features / data.session_features	head / conditions_v1.npz	off /	(3.6)
	condition projection size	model.session_head_dim	32		(3.6)
	output init scale	model.init_scale	0.001		
target	task	model.task	regression		(3.2)
loss	regression loss / δ	loss.regression_loss / loss.huber_delta	huber / 1		(3.4)
augmentation	mixup α (on)	mixup.alpha / mixup.enabled	0.4 / True		(3.3)
optimizer	AdamW lr / batch	optimizer.lr / trainer.batch_size	0.0002 / 128		
	weight decay / layer decay	optimizer.weight_decay / optimizer.layer_decay	0.05 / 0.65		
schedule	epochs / warmup	trainer.epochs / optimizer.warmup_epochs	10 / 3	5 / 1	
	min lr	optimizer.min_lr	1e-06		
selection	checkpoint	–	lowest val MAE (recording level)		
scoring	window aggregation / case unit	evaluation.agg_windows / evaluation.agg_case_by	mean / recording		(3.7)

Table 3.2. The three cross-validated configurations

Every configuration field on which `cvbase`, `cvctrl` and `cvaug` disagree (`labram/eval/research_summary.py::config_diff`, bookkeeping fields excluded).

field	cvbase	cvctrl	cvaug
channel_drop.consistency_weight	0	0	0.1
channel_drop.prob	0	0	0.2
data.session_features		conditions_v1.npz	conditions_v1.npz
model.session_features	off	head	head
model.use_mean_pooling	True	False	False
optimizer.warmup_epochs	1	3	3
trainer.epochs	5	10	10

Table 3.3. Results of the best experiment by evaluation set and cohort

`cvctrl` (best overall) and `cvbase` (best EEG-only), mean ± sample sd over the 5 fold models of the MAE (Eq. (0.1)) and the mean R^2 (Eq. (0.2)), per evaluation set and status cohort:

- `CV val`, `CV test`: the fold's own held-out folds, TUAB and TUEG recordings together;
- `TUEG test`: the TUEG recordings of each fold's test fold;
- `TUAB test`: the official test set, $n = 274$.

R^2 is relative to each cohort's own age spread, so it does not compare across cohorts. The last row gives the median ensemble (Eq. (3.8)) on the TUAB test set when the per-recording predictions are available. For reference, the best published TUAB normal-recording result is MAE 6.60, R^2 0.73 (Gemein et al. 2024, trained on TUAB normal recordings only, on its own patient-wise re-split of TUAB).

set	cohort	cvctrl MAE	cvctrl R²	cvbase MAE	cvbase R²
CV val	all	8.09 ± 0.12	0.64	8.16 ± 0.11	0.64
	normal	7.33 ± 0.20	0.68	7.45 ± 0.17	0.68
	abnormal	8.49 ± 0.14	0.60	8.57 ± 0.13	0.59
CV test	all	8.08 ± 0.13	0.64	8.18 ± 0.15	0.63
	normal	7.39 ± 0.18	0.68	7.47 ± 0.14	0.67
	abnormal	8.46 ± 0.11	0.60	8.60 ± 0.15	0.59
TUEG test	all	8.17 ± 0.15	0.63	8.27 ± 0.15	0.62
	normal	7.52 ± 0.19	0.67	7.59 ± 0.16	0.67
	abnormal	8.50 ± 0.10	0.59	8.65 ± 0.13	0.58
TUAB test	all	7.28 ± 0.14	0.71	7.49 ± 0.09	0.70
	normal	5.53 ± 0.08	0.80	5.81 ± 0.10	0.79
	abnormal	9.37 ± 0.25	0.52	9.50 ± 0.11	0.51
TUAB test, ensemble	all	7.10	0.72	7.44	0.70
	normal	5.31	0.82	5.75	0.80
	abnormal	9.23	0.53	9.46	0.52

Table 3.4. Predicted versus true age by subcohort and status

The three cross-validated models, scored per recording on three sets of recordings (`rs.prediction_sets`, from the per-recording CV predictions):

- `TUAB test, ensemble`: the 274 TUAB test recordings, predicted by the median of the 5 fold models (Eq. (3.8)).
- `TUAB out-of-fold`: the TUAB recordings of the CV pool (TUAB train + val), each predicted by the one fold model that never saw its patient.
- `TUEG out-of-fold`: the same for the pool's TUEG recordings that are not in TUAB.

`status` is the recording's normal / abnormal / no-label status (Eq. (1.2)), and `all` is every recording of the set. Per model, the columns are:

- `MAE` (Eq. (0.1), years);
- `bias` = mean(predicted − true) (Eq. (0.3), years): the signed drift. It is negative when the model predicts too young and positive when too old, so closer to 0 is better;
- R^2 (Eq. (0.2));
- the bias slope β (Eq. (0.5)).

All come from `rs.prediction_metrics`. Needs the per-recording CV predictions and the TUH index.

			cvbase				cvctrl				cvaug			
		n	MAE	bias	R²	β	MAE	bias	R²	β	MAE	bias	R²	β
set	status													
TUAB test, ensemble	all	274	7.44	-1.21	0.70	-0.37	7.10	-0.77	0.72	-0.35	7.23	-0.76	0.72	-0.34
	normal	149	5.75	-0.56	0.80	-0.30	5.31	-0.58	0.82	-0.29	5.43	-0.67	0.81	-0.29
	abnormal	125	9.46	-1.99	0.52	-0.50	9.23	-1.00	0.53	-0.50	9.37	-0.88	0.54	-0.49
TUAB out-of-fold	all	2350	7.18	-0.96	0.70	-0.35	7.11	-0.45	0.71	-0.34	7.12	-0.86	0.71	-0.33
	normal	1258	6.55	-0.85	0.70	-0.38	6.37	-0.61	0.71	-0.34	6.41	-1.15	0.71	-0.35
	abnormal	1092	7.91	-1.08	0.63	-0.41	7.97	-0.27	0.63	-0.41	7.93	-0.52	0.63	-0.41
TUEG out-of-fold	all	26685	8.27	-0.29	0.62	-0.42	8.17	0.02	0.63	-0.39	8.14	-0.05	0.63	-0.39
	normal	10197	7.59	-0.24	0.67	-0.38	7.51	-0.30	0.68	-0.36	7.48	-0.45	0.68	-0.36
	abnormal	14298	8.65	-0.48	0.58	-0.45	8.50	0.20	0.59	-0.43	8.50	0.16	0.59	-0.43
	no label	2190	8.99	0.78	0.51	-0.50	9.07	0.32	0.51	-0.50	8.94	0.38	0.51	-0.49

Figure 3.1. Predicted versus true age, per model and subcohort

Predicted age $\hat{y}_i$ against true age y_i (years), one point per recording; the data of Table 3.4. One row per cross-validated model, and two subcohorts per row:

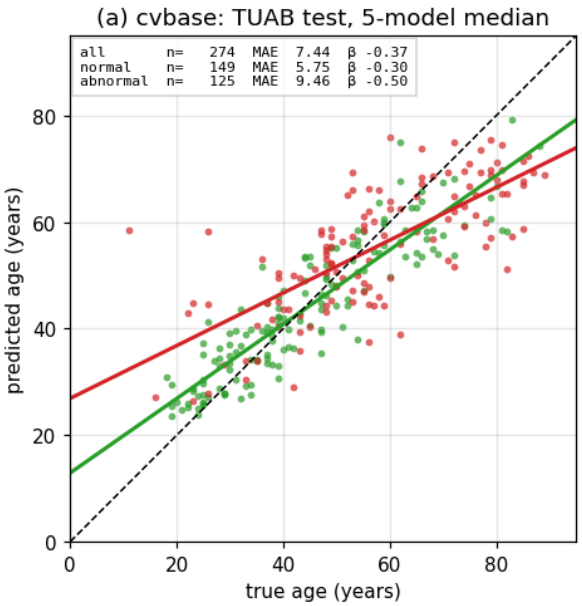
- left column: the 274 TUAB test recordings, predicted by the 5-model median ensemble;
- right column: the TUEG out-of-fold recordings (n = 26,685), each predicted by the fold model that held it out.

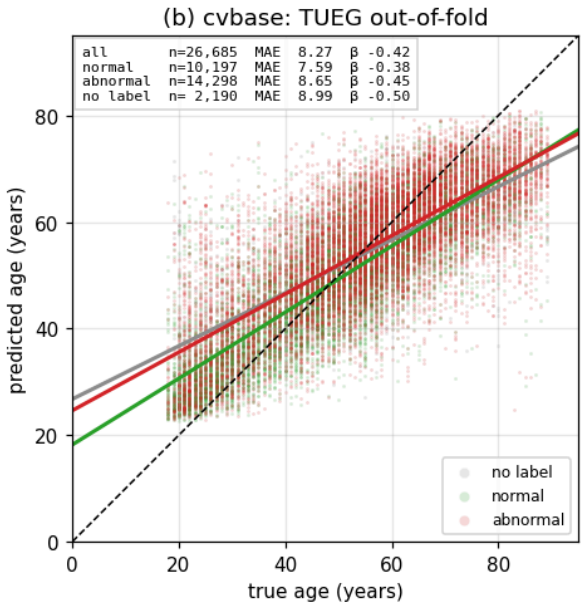
The panels are:

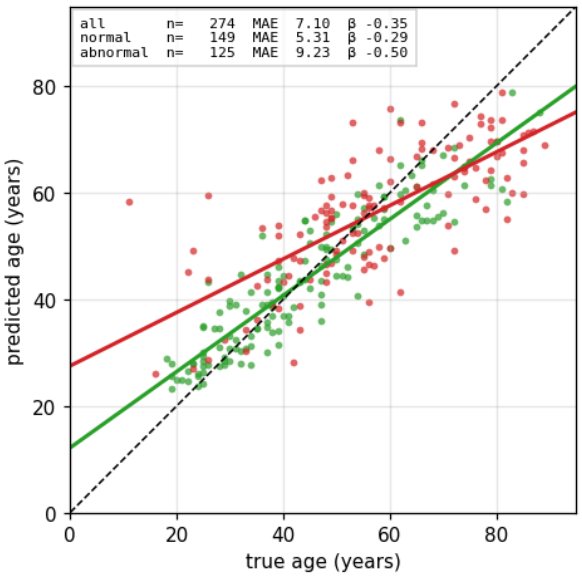
- (a), (b) `cvbase`, the best EEG-only model;
- (c), (d) `cvctrl`, the best overall;
- (e), (f) `cvaug`.

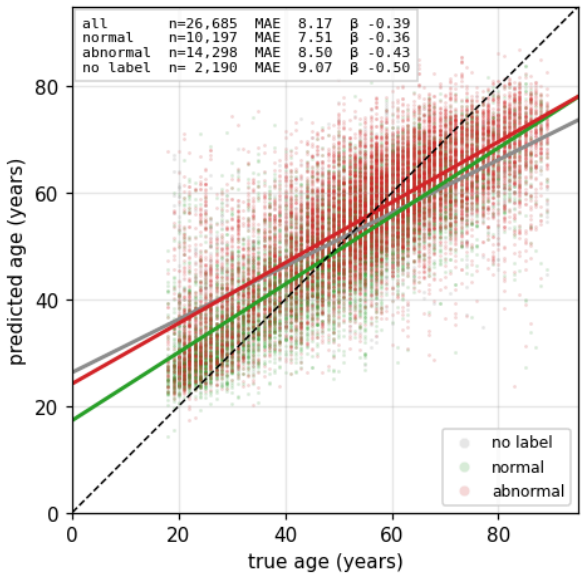
Colour = status (Eq. (1.2)): green normal, red abnormal, grey no label. The no-label group occurs only in TUEG.

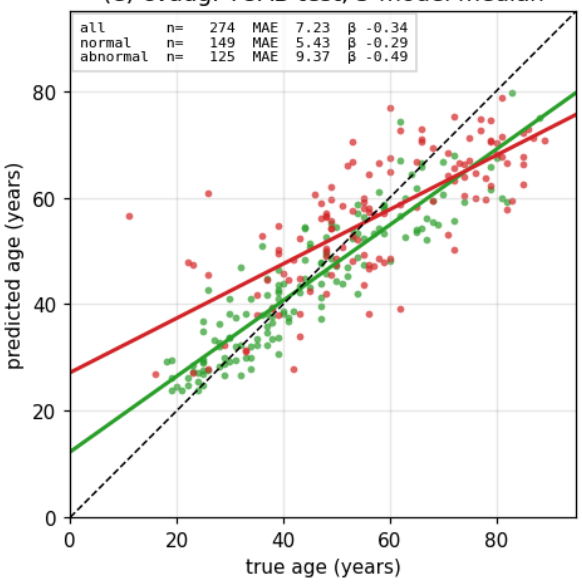
- **Dashed black diagonal:** a perfect prediction, $\hat{y} = y$.
- **Coloured lines:** least-squares fits of $\hat{y}$ on y per status. A fit flatter than the diagonal is the regression to the mean that β measures: the fit's slope is $1 + \beta$ (Eq. (0.5)).
- **Box in each panel:** n, MAE and β per status, and for all recordings together.

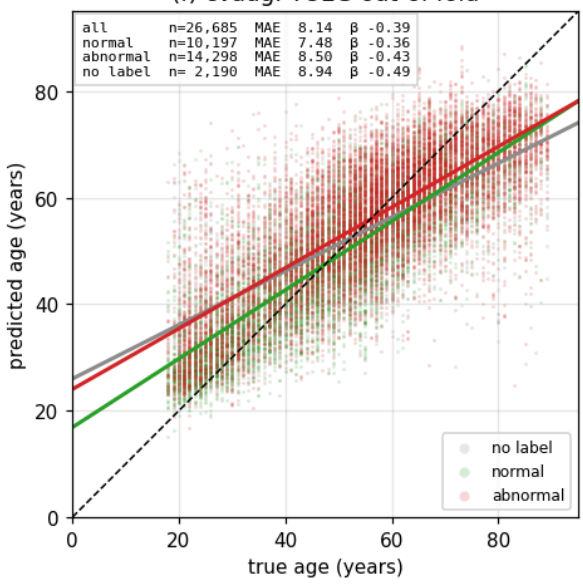

(a) cvbase: TUAB test, 5-model median
all n= 274 MAE 7.44 β -0.37
normal n= 149 MAE 5.75 β -0.30
abnormal n= 125 MAE 9.46 β -0.50
predicted age (years)
true age (years)


(b) cvbase: TUEG out-of-fold
all n=26,685 MAE 8.27 β -0.42
normal n=10,197 MAE 7.59 β -0.38
abnormal n=14,298 MAE 8.65 β -0.45
no label n= 2,190 MAE 8.99 β -0.50
no label
normal
abnormal
predicted age (years)
true age (years)


(c) cvctrl: TUAB test, 5-model median
all n= 274 MAE 7.10 β -0.35
normal n= 149 MAE 5.31 β -0.29
abnormal n= 125 MAE 9.23 β -0.50
predicted age (years)
true age (years)


(d) cvctrl: TUEG out-of-fold
all n=26,685 MAE 8.17 β -0.39
normal n=10,197 MAE 7.51 β -0.36
abnormal n=14,298 MAE 8.50 β -0.43
no label n= 2,190 MAE 9.07 β -0.50
no label
normal
abnormal
predicted age (years)
true age (years)


(e) cvaug: TUAB test, 5-model median
all n= 274 MAE 7.23 β -0.34
normal n= 149 MAE 5.43 β -0.29
abnormal n= 125 MAE 9.37 β -0.49
predicted age (years)
true age (years)


(f) cvaug: TUEG out-of-fold
all n=26,685 MAE 8.14 β -0.39
normal n=10,197 MAE 7.48 β -0.36
abnormal n=14,298 MAE 8.50 β -0.43
no label n= 2,190 MAE 8.94 β -0.49
no label
normal
abnormal
predicted age (years)
true age (years)

Reading Table 3.4 and Figure 3.1.

- **Abnormal recordings are harder in every set.**
  - On TUAB test, their MAE is 3.7–3.9 years above the normal recordings'.
  - In TUEG out-of-fold the gap is about 1 year (0.99–1.06). The no-label recordings are the hardest there (MAE 8.9–9.1).
- **Every model regresses to the mean** ($\beta \approx -0.3$ to -0.5). The fit lines in Figure 3.1 are visibly flatter than the diagonal: old patients are predicted too young and young ones too old. The effect is strongest for the abnormal and no-label recordings ($\beta \approx -0.43$ to -0.50).
- **Bias.** The overall drift is small: within ±1.2 years on TUAB test and ±0.3 on TUEG. But the EEG-only `cvbase` predicts abnormal TUAB test recordings 2.0 years too young on average, against 0.9–1.0 for `cvctrl` / `cvaug`.
- **TUEG is harder than TUAB.** TUEG out-of-fold MAE is 8.1–8.3, against 7.1–7.2 for the TUAB recordings of the same pool. Its normal recordings are worse too: 7.5–7.6 against 6.4–6.6. TUEG's status comes from report text rather than TUAB's curated folders. It also covers a broad clinical population: 38 % of its recordings in the clinical sheet list a sedating drug (Table 1.6).
- **cvctrl and cvaug beat cvbase on the TUAB test ensemble** by 0.2–0.35 years. Their gain is larger on normal recordings (0.3–0.4 years) than on abnormal ones (0.1–0.2).

Table 3.5. TUAB test MAE before and after age-bias correction, and published / classical methods

Every other number in this notebook is before bias correction. That includes the MAEs of Sections 0–5 and Figure 2.1. They score the model's raw prediction $\hat{y}_i$. Brain-age studies often also report a *bias-corrected* error. Gemein et al. 2024 does: its 6.60 / 12.85 are after correction, and its raw normal MAE was 6.48. This table puts our CV models on that footing, and lists the other methods evaluated on the TUAB test set.

Where the correction is fitted: never on the test set. Per experiment, the brain-age gap is fitted as a polynomial p of the true age on the experiment's own **out-of-fold** predictions. In the 5-fold CV, each recording of the CV pool (TUAB train + val, TUEG train + val) is predicted by the one fold model that held it out. These are therefore validation predictions: 29,035 recordings of 13,550 patients. The check below the table confirms that none of them is a TUAB test recording and none shares a patient with the TUAB test set. The fitted correction is then applied to the TUAB test predictions (`rs.bias_corrected_test`):

$$e_i = \hat{y}_i - y_i \approx p(y_i), \qquad \hat{y}_i^c = \hat{y}_i - p(y_i) \tag{3.9}$$

- **Degree:** p is linear or quadratic. Gemein et al. use a quadratic.
- **Fit data:** either all out-of-fold recordings, or only the normal ones (as Gemein et al., whose model only ever saw normal recordings).

- **Caveat:** Eq. (3.9) is *applied* with each test recording's **true age** y_i. The corrected error therefore measures the brain-age gap after removing its age trend. It is not the error of an age estimate.

The **deployable** alternative (Cole et al.) uses only the prediction. Fit $\hat{y} \approx a\,y + c$ on the same out-of-fold recordings and invert it (`rs.cole_corrected_test`):

$$\hat{y}'_i = (\hat{y}_i - c) \,/\, a \tag{3.10}$$

Columns. TUAB test MAE (Eq. (0.1), years) per model and status:

- `raw` : no correction (as in Table 3.4);
- `cole, normal` : Eq. (3.10), fitted on the normal out-of-fold recordings. Deployable;
- `lin, all` / `quad, all` : Eq. (3.9), linear or quadratic, fitted on all out-of-fold recordings;
- `quad, normal` : quadratic, fitted on the normal out-of-fold recordings only;
- `bias, quad normal` : the mean gap (Eq. (0.3)) after that correction. A positive value means the recordings look older than they are.

Other methods, with each source's own numbers; "–" means the source does not report it:

- **Gemein et al. 2024:** a TCN ensemble trained on TUEG normal recordings, scored on TUAB's evaluation set. Raw normal 6.48; after their quadratic correction, 6.60 normal and 12.85 abnormal.
- **Engemann et al. 2022:** the M/EEG brain-age benchmark. Best model 7.75 on TUAB normal recordings, raw, 10-fold CV on the TUAB normal training recordings (not the official evaluation set).
- **Handcrafted features + ridge** (this repo, `notebooks/eeg_feature_extraction_tuab.ipynb` §10). This is the classical approach of Engemann et al.: filterbank-Riemann, spectral, wavelet and other EEG features with a ridge regression. It is scored raw on **the same 274 TUAB test recordings**, and trained either on TUAB normal recordings only (the standard brain-age setup) or on all TUAB recordings.

model	status	raw	cole, normal	lin, all	quad, all	quad, normal	bias, quad normal
cvbase	all	7.44	9.09	6.06	5.92	5.61	-0.05
	normal	5.75	7.77	5.65	5.47	4.77	-1.63
	abnormal	9.46	10.67	6.56	6.45	6.60	1.84
cvctrl	all	7.10	8.86	6.03	5.85	5.59	0.39
	normal	5.31	7.35	5.53	5.37	4.62	-1.55
	abnormal	9.23	10.67	6.61	6.43	6.74	2.70
cvaug	all	7.23	8.81	6.05	5.92	5.63	0.57
	normal	5.43	7.16	5.54	5.39	4.55	-1.42
	abnormal	9.37	10.78	6.67	6.55	6.91	2.96
Gemein et al. 2024 (TCN, TUEG normal)	normal	6.48	–	–	–	6.60	–
	abnormal	–	–	–	–	12.85	–
Engemann et al. 2022 (TUAB normal, 10-fold CV)	normal	7.75	–	–	–	–	–
features + ridge, trained on TUAB normal	all	9.70	–	–	–	–	–
	normal	7.21	–	–	–	–	–
	abnormal	12.67	–	–	–	–	–
features + ridge, trained on all TUAB	all	8.53	–	–	–	–	–
	normal	6.95	–	–	–	–	–
	abnormal	10.41	–	–	–	–	–

```
Fit-set check (bias correction):
  cvaug: fitted on 29,035 out-of-fold recordings of 13,550 patients; 0 of them TUAB tes
t, 0 sharing a test patient
  cvbase: fitted on 29,035 out-of-fold recordings of 13,550 patients; 0 of them TUAB te
st, 0 sharing a test patient
  cvctrl: fitted on 29,035 out-of-fold recordings of 13,550 patients; 0 of them TUAB te
st, 0 sharing a test patient
```

Reading Table 3.5.

- **No test leakage.** Every correction is fitted on out-of-fold (validation) predictions. The check above finds 0 TUAB test recordings and 0 shared patients in the fit set.
- **The deployable correction makes the error worse** (`cole, normal`): 7.1–7.4 raw becomes 8.8–9.1 overall. Spreading the predictions out to remove the bias also spreads out their noise. The raw prediction is the best age estimate; the corrected numbers are for brain-age-gap analysis only.
- **Normal recordings, gap correction.** Fitting on all recordings changes little (5.3–5.8 raw, 5.4–5.7 corrected). With Gemein's choice, a quadratic fitted on normal recordings, the normal MAE falls to 4.6–4.8 years, against Gemein's 6.60 after the same kind of correction.
- **Abnormal recordings.** The gap correction removes most of their error: 9.2–9.5 raw becomes 6.4–7.0. Much of the raw abnormal error is regression to the mean ($\beta \approx -0.5$, Table 3.4), and Eq. (3.9) undoes it with the true age.
- **Abnormal recordings look older.** After the normal-fit correction they keep a positive gap of +1.8 to +3.0 years (`bias, quad normal`), and normal recordings a negative one. This is the brain-age-gap effect Gemein et al. report: −4 years between normal and pathological subjects.
- **Against the other methods, raw on the same test set.** The EEG-only `cvbase` scores 7.44 all / 5.75 normal / 9.46 abnormal. The classical feature-ridge model scores 8.53 / 6.95 / 10.41 when trained on all of TUAB, and 9.70 / 7.21 / 12.67 trained on normal recordings only. Engemann et al.'s best is 7.75 on normal recordings (CV, not the evaluation set). The foundation model trained on TUAB + TUEG is better on every cohort. Against the feature model trained on all of TUAB it is 0.95–1.2 years ahead (smallest on abnormal recordings). Against the normal-only feature model it is 1.5–3.2 years ahead, most on abnormal recordings, where normal-only models fail (12.67 here, 12.85 for Gemein et al.).
- **Ranking.** All three of our models are within 0.35 years of each other after correction.

Table 3.6. Agreement between experiments on the TUAB test set

How similar are the models' predictions, recording by recording? The comparison covers two kinds of pair:

- every pair of experiment ensembles (cvbase / cvctrl / cvaug);
- as a reference, the 10 pairs of fold models *within* each experiment. These pairs share the configuration but differ in training data and seed.

For two predictors a and b on the same recordings, with errors $e_i^a = \hat{y}_i^a - y_i$ (`labram/eval/cv_compare.py::pairwise`):

$$\mathrm{agree}_\tau = \frac{1}{n}\sum_i 1\big[|\hat{y}_i^a - \hat{y}_i^b| \le \tau\big], \qquad \mathrm{sign} = \frac{1}{n}\sum_i 1\big[\operatorname{sgn} e_i^a = \operatorname{sgn} e_i^b\big] \tag{3.11}$$

Columns, per status (all 274, normal 149, abnormal 125):

- `r_pred` , `r_err` : the Pearson correlation of the two models' predictions and of their errors. `r_err` near 1 means they make the same mistakes on the same recordings.
- `agree_2y` , `agree_5y` : agree_τ for τ = 2 and 5 years.
- `sign_agree` : Eq. (3.11, right). The share of recordings where both models err in the same direction (both predict too old, or both too young).
- `dMAE` : $\mathrm{MAE}_a - \mathrm{MAE}_b$ (Eq. (0.4)) for an ensemble pair; the mean $|\Delta\mathrm{MAE}|$ over the 10 pairs for a fold-model row.

status	pair	r_pred	r_err	agree_2y	agree_5y	sign_agree	dMAE
all	cvbase vs cvctrl (ensembles)	0.987	0.974	0.693	0.960	0.912	0.340
	cvbase vs cvaug (ensembles)	0.989	0.977	0.701	0.978	0.912	0.214
	cvctrl vs cvaug (ensembles)	0.994	0.987	0.821	0.989	0.920	-0.126
	cvbase: fold model vs fold model (mean of 10 pairs)	0.982	0.966	0.668	0.934	0.931	0.107
	cvctrl: fold model vs fold model (mean of 10 pairs)	0.969	0.936	0.532	0.876	0.889	0.177
	cvaug: fold model vs fold model (mean of 10 pairs)	0.972	0.939	0.520	0.886	0.872	0.208
normal	cvbase vs cvctrl (ensembles)	0.992	0.975	0.805	0.980	0.893	0.439
	cvbase vs cvaug (ensembles)	0.992	0.976	0.819	0.993	0.899	0.319
	cvctrl vs cvaug (ensembles)	0.995	0.984	0.899	1.000	0.886	-0.120
	cvbase: fold model vs fold model (mean of 10 pairs)	0.989	0.969	0.759	0.982	0.914	0.108
	cvctrl: fold model vs fold model (mean of 10 pairs)	0.981	0.939	0.621	0.938	0.872	0.101
	cvaug: fold model vs fold model (mean of 10 pairs)	0.985	0.949	0.580	0.940	0.851	0.209
abnormal	cvbase vs cvctrl (ensembles)	0.976	0.976	0.560	0.936	0.936	0.222
	cvbase vs cvaug (ensembles)	0.980	0.980	0.560	0.960	0.928	0.088
	cvctrl vs cvaug (ensembles)	0.988	0.988	0.728	0.976	0.960	-0.134
	cvbase: fold model vs fold model (mean of 10 pairs)	0.963	0.964	0.558	0.878	0.950	0.133
	cvctrl: fold model vs fold model (mean of 10 pairs)	0.936	0.936	0.427	0.802	0.909	0.311
	cvaug: fold model vs fold model (mean of 10 pairs)	0.941	0.937	0.450	0.822	0.898	0.214

Figure 3.2. Error correlation between all TUAB-test predictors

The Pearson correlation of the errors $e_i = \hat{y}_i - y_i$ between every pair of the 18 predictors of the TUAB test set: each experiment's 5 fold models (`f0` – `f4`) and its median ensemble (`ens`).

- (a) The 149 normal recordings.
- (b) The 125 abnormal recordings.

The colour scale runs from 0.85 (lighter) to 1 (darker). The thin white lines separate the three experiments.

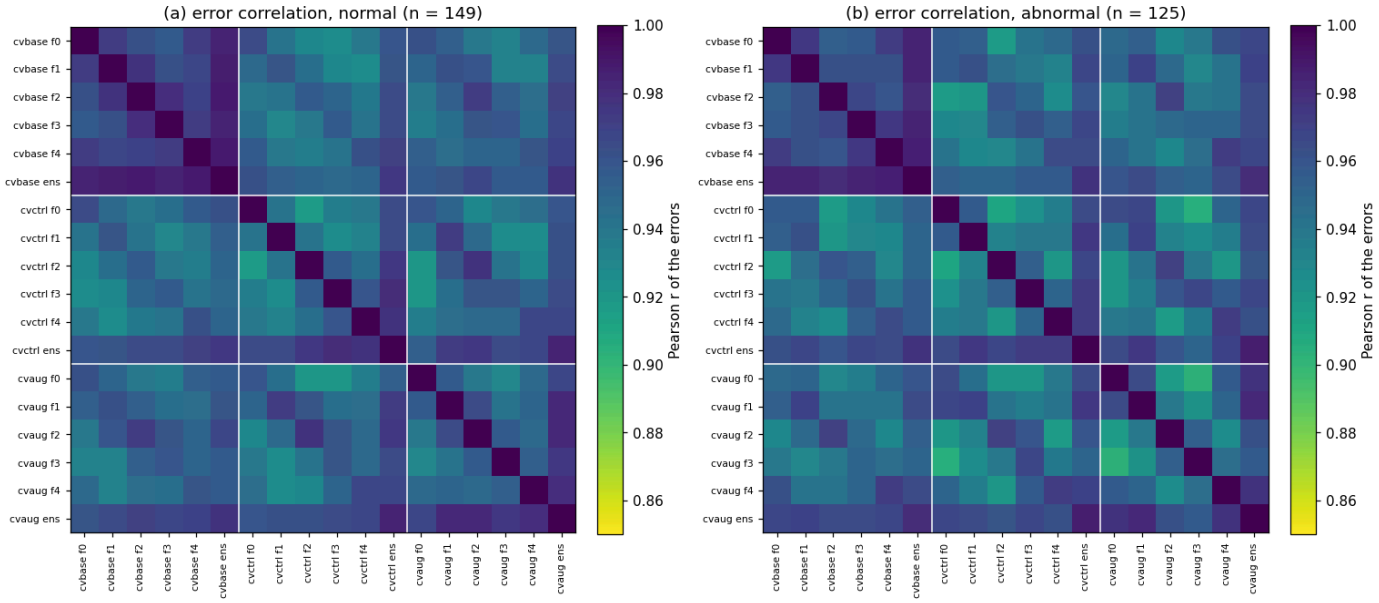
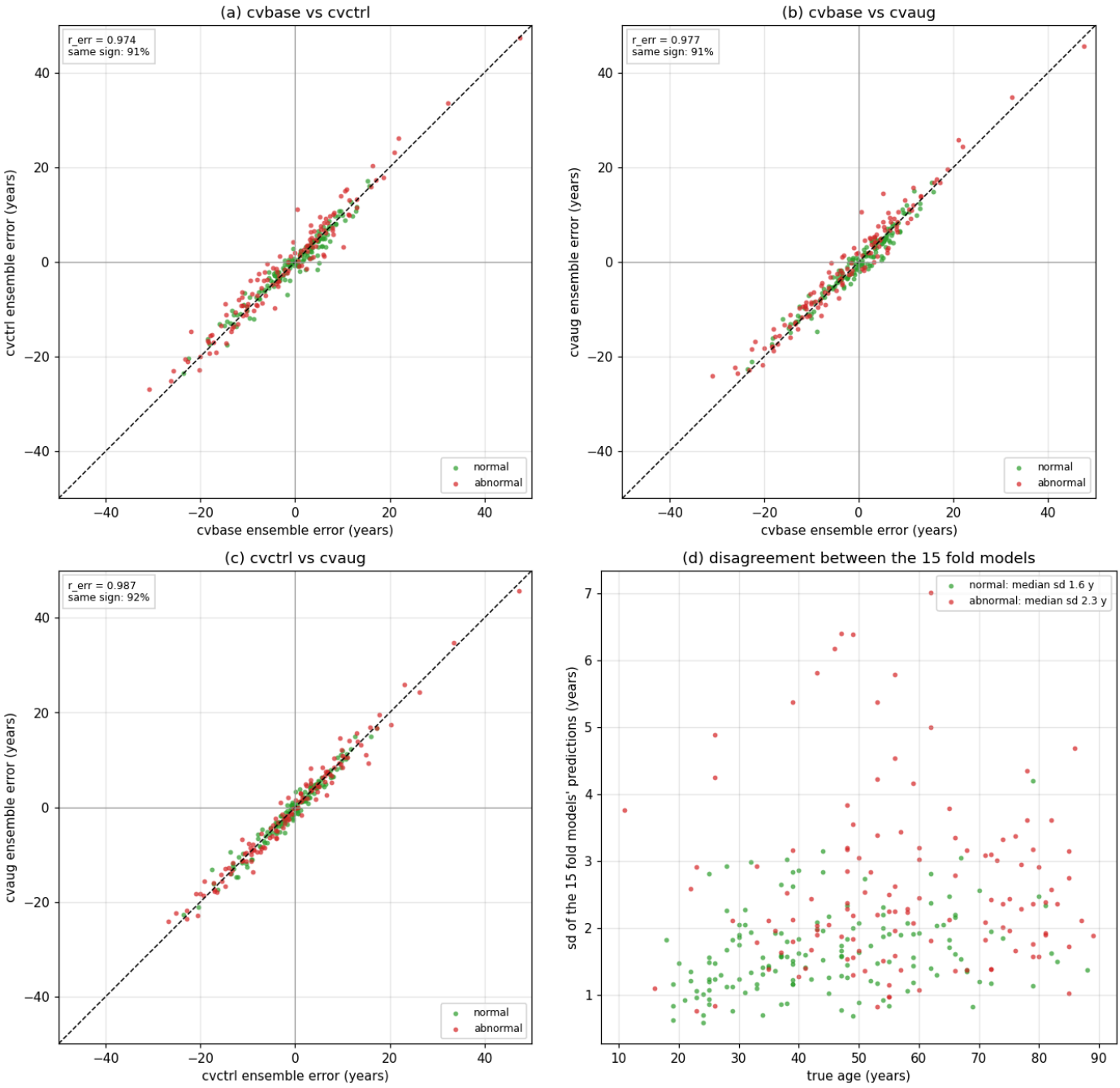


Figure 3.3. Per-recording agreement of the three experiments

TUAB test recordings, coloured by status: green normal, red abnormal. Errors $e = \hat{y} - y$ are in years.

- (a)–(c) The error of one experiment's ensemble against another's, for each pair of experiments. A point on the dashed diagonal means both models made the same error. A point in the upper-right or lower-left quadrant means both erred in the same direction.
- (d) **Disagreement per recording:** the sample standard deviation of the 15 fold models' predictions (3 experiments × 5 folds), against the true age.



Reading Table 3.6 and Figures 3.2 and 3.3.

- **The three experiments make almost the same errors.** Their ensembles' errors correlate at r = 0.974–0.987 (Figure 3.2, Figure 3.3(a)–(c)). On 91–92 % of the recordings both err in the same direction, and on 96–99 % they predict within 5 years of each other. The differences in MAE between them (Table 2.4) come from small shifts, not from different recordings going wrong.
- **The ensembles agree more than single fold models do.** The ensemble pairs agree more closely across experiments (r_err 0.97–0.99, within 2 years on 69–82 % of recordings) than two fold models of the same experiment (r_err 0.94–0.97, 52–67 %). Averaging five models removes much of the fold-to-fold noise, which is why the ensemble beats a single model (Table 5.2, C3).

- **Abnormal recordings: shared errors, more disagreement.** The models agree on the error direction more often on abnormal recordings: 93–96 % same sign, against 89–90 % on normal ones. But their predictions spread more there, with fewer abnormal recordings within 2 years (56–73 %, against 81–90 %). The fold models also disagree more on abnormal recordings (Figure 3.3(d)).
- **The largest errors are shared.** In Figure 3.3(a)–(c), the points far from the origin lie near the diagonal: the recordings that one model gets badly wrong, all models get badly wrong. Of the 20 largest `cvctrl` test errors, 16 are abnormal recordings (46 % of the test set is abnormal), and 17 are patients under 30 or over 70. These recordings are the main lever left for improving the MAE.

4. Changes versus the LaBraM recipe

The reference is **LaBraM as published for TUAB**, `labram/configs/defaults/finetune_tuab_paper.json`:

- LaBraM-base fine-tuned for normal/abnormal classification, the recipe of the paper and the README;
- lr $5 \cdot 10^{-4}$, batch 64, 50 epochs with 5 warmup, layer decay 0.65, drop path 0.1;
- no input re-referencing;
- a 50 Hz notch in preprocessing (`dataset_maker/make_TUAB.py`).

Table 4.1 lists every configuration field the best configuration changes. Table 4.2 lists the changes that live in the data or the code rather than the configuration. Section 5 then tests which of them moved the error.

Table 4.1. Configuration changes: LaBraM TUAB recipe → best configuration

Every field on which `finetune_tuab_paper.json` and `cvctrl` (Table 3.1) differ, computed by `labram/eval/research_summary.py::config_diff`. Excluded:

- bookkeeping fields (paths, logging, infrastructure);
- fields a regression run never reads (`INERT_FOR_REGRESSION`);
- fields of a section that is off in both.

Columns:

- `group` : the part of the pipeline;
- `meaning` : what the change does;
- `evidence` : the rows of Table 5.2 that test it, where one does.

group	field	LaBraM recipe	best (cvctrl)	meaning	evidence
augmentation	**mixup.enabled**	False	True	mixup α 0.4 (Eq. (3.3))	P3
channels	**channel_drop.channels**	[]	["T1", "T2"]	T1/T2 are the channels that may be missing	Q14
	channel_drop.consistency_weight	1	0	0: no consistency loss in cvctrl	Q17, C2
	channel_drop.enabled	False	True	per-sample handling of missing T1/T2 (needed for cohort v3)	Q14
	channel_drop.max_drop	1	2	up to both of T1/T2	Q14
	channel_drop.prob	0.2	0	0: no channel-dropout augmentation in cvctrl	Q17, C2
data	**data.data_format**	pickle	npy	storage only: one float32 npy per recording, windows bit-identical to the pickles	
head	**data.session_features**		conditions_v1.npz	the condition vectors' file (conditions_v1.npz)	Q12, C1
	model.session_features	off	head	status + pathology vector concatenated to the feature (Eq. (3.6))	Q12, C1
	model.use_mean_pooling	True	False	head on the class token instead of the mean patch token	P22, Q13, C1
input	**labram_plus.enabled**	False	True	turns on the LaBraM++ input block (Eq. (3.1))	P1
	labram_plus.z_score_patches	True	False	LaBraM++ block with CAR only, no per-patch z-scoring	P1
optimizer	**optimizer.lr**	0.0005	0.0002	2e-4 at batch 128 = 1e-4 at batch 64, linearly scaled	Q14
	trainer.batch_size	64	128	batch 128 (speed; lr scaled with it)	Q14
protocol	**cross_validation.enabled**	False	True	5-fold patient-grouped cross-validation	C3
schedule	**optimizer.warmup_epochs**	5	3	3 warmup epochs	P2, Q6, C1
	trainer.epochs	50	10	10 epochs instead of 50 (best epoch is early)	P2, Q6, C1
scoring	**evaluation.agg_windows**	none	mean	one prediction per recording = mean of its windows (Eq. (3.7))	
speed	**model.fused_attention**	False	True	fused attention kernel, numerically the same model	
task	**data.dataset**	TUAB	TUAB_AGE	age regression on TUH (TUAB_AGE loader) instead of normal/abnormal	
	model.task	classification	regression	scalar regression head (Huber on z-scored age, Eqs. (3.2), (3.4))	
windows	**data.eval_minutes**	0	5	score each recording on its first 5 minutes after the trim	P26
	data.train_minutes_per_epoch	0	5	5 minutes per recording per epoch, redrawn each epoch (speed)	Q14
	data.trim_end_sec	0	60	drop the last 60 s of every recording	
	data.trim_start_sec	0	60	drop the first 60 s of every recording	

Table 4.2. Changes outside the configuration

Changes in the training data, the preprocessing, the split and the code, with the Table 5.2 rows that test them.

change	LaBraM recipe	this study	evidence
training data	TUAB train, 2,145 recordings (≈1.6k patients)	TUAB + TUEG, 25,613 recordings (≈12k patients, `all_both_v3`)	Q1, Q3
mains interference	50 Hz notch (TUH is recorded at 60 Hz, so the line stays in)	50 + 60 Hz spectrum interpolation	L1
channels	23 fixed channels	23; TUEG recordings missing only T1/T2 kept, with those channels left out per sample	Q14
validation split	TUAB's train split, which leaks 16 patients between train and val	patient-disjoint, seeded; one TUEG patient split shared by all cohorts; TUAB-test patients removed from TUEG	Q11
label	normal/abnormal	age from the EDF header (18–89 in the TUH cohorts)	–
unit of evaluation	10 s window	recording (Eq. (3.7)), first 5 minutes after the trim	P26
per-patient cap	none	at most 10 train/val recordings per patient	–
reported model	one run	median of 5 fold models (Eq. (3.8))	C3
reproducibility	ClearML silently reseeded every tracked run to 1337	fixed: `distributed.seed` is honoured (commit `7b9a5fe`)	–

5. Which changes mattered

Noise floor. Two runs of the same configuration with different seeds do not give the same MAE. The line-removal A/B study ran each of 3 preprocessing recipes r with 3 seeds s, and each run is read at its best-validation epoch. The pooled within-recipe sample standard deviation of a metric m (`labram/eval/research_summary.py::seed_noise`) is

$$\hat{\sigma}_m^2 = \frac{1}{N-R} \sum_r \sum_s \left(m_{r,s} - \bar{m}_r\right)^2 \tag{5.1}$$

with $N = 9$ runs and $R = 3$ recipes.

Significance rule. Each row of Table 5.2 compares a baseline with a variant that changes one thing (or a named bundle), using whichever evidence exists:

- **One run each** (phases 1 and 2). The difference of two single runs has standard deviation $\sqrt{2}\,\hat{\sigma}$, so a change counts only if (`rs.pair_threshold`)

$$|\Delta\mathrm{MAE}| > \tau = 1.96\,\sqrt{2}\,\hat{\sigma}_{\mathrm{test}}. \tag{5.2}$$

- **Three seeds each** (line removal, L1/L2):

$$|\bar{\Delta}| > 1.96\,\hat{\sigma}_{\mathrm{test}}\,\sqrt{2/3}. \tag{5.3}$$

- **Five paired folds** (CV, C1/C2). Fold k gives Δ_k on the same TUAB test recordings, and the 95 % interval is

$$\bar{\Delta} \pm t_{0.975,\,4}\,\frac{s_\Delta}{\sqrt{5}}, \qquad t_{0.975,\,4} = 2.776, \tag{5.4}$$

  where $\bar{\Delta}$ and s_Δ are the mean and sample sd of the Δ_k. The change counts if the interval excludes 0.

- **Ensemble vs single model** (C3). A paired bootstrap over recordings (2,000 resamples, `cv_compare.py::bootstrap_delta_mae`) of the ensemble's absolute error against the fold models' mean absolute error. The change counts if its 95 % interval excludes 0.

The verdict is `improves` / `hurts` / `n.s.` (`rs.verdict`). It is judged on the TUAB test MAE of all 274 recordings, the one set every run shares; the val and cohort columns are context.

Limits.

- $\hat{\sigma}$ is measured on one configuration, the E2 recipe on TUAB (Table 5.1), and assumed to hold for the others.
- A single-split row with $|\Delta|$ just under τ is not evidence of no effect, only of no detectable one.
- Several rows change more than one thing at once; they are marked *bundle*.

Table 5.1. Seed-to-seed spread

The model. All 9 runs train the same model with the same recipe, **E2**: scenario D (`finetune_tuab_age.json`) on the short schedule. Read from each run's ClearML configuration:

- **Encoder:** LaBraM-base (`labram_base_patch200_200` , 12 transformer blocks), initialized from the released `labram-base.pth` , all layers fine-tuned.
- **Input and head:** CAR input (Eq. (3.1)), mean patch-token pooling and a linear head.
- **Loss and optimizer:** Huber $\delta = 1$ on the z-scored age (Eq. (2.3)), no mixup. AdamW with lr 10^{-4}, batch 64, weight decay 0.05, layer decay 0.65 and drop path 0.1.
- **Schedule and evaluation:** 4 epochs with 1 warmup, scored on the first 5 minutes of each recording after the 60 s trims.

What varies. The data is TUAB v3.0.1, on the same recordings and the same subject-disjoint split as phase 1 (Table 1.2). Only two things change:

- **Preprocessing:** the mains-interference removal, 3 variants. A is the 50 Hz notch (LaBraM as published), B a 60 Hz notch, C 50 + 60 Hz spectrum interpolation.
- **Seed:** 3 seeds per variant, which change the head initialization, the batch order, dropout and drop path. Seed 0 ran on a local g5.2xlarge; seeds 1 and 2 ran as SageMaker jobs on ml.g5.2xlarge (`scripts/submit_line_removal_ab.sh`).

Within a variant, the three runs differ only by the seed, which is what Eq. (5.1) pools. Each run is read at its best-validation epoch (`docs/data/line_removal_ab.json`). The table then gives the pooled standard deviation of Eq. (5.1) per metric and the single-pair threshold τ of Eq. (5.2).

Columns:

- `val_mae` : the TUAB validation split.
- `test_mae` , `test_normal_mae` , `test_abnormal_mae` : TUAB test, overall and per cohort.

variant	seed	epoch	val_mae	test_mae	test_normal_mae	test_abnormal_mae
A_notch50	0	3	7.52	8.43	6.62	10.59
	1	3	7.66	8.40	6.60	10.55
	2	3	7.64	8.24	6.41	10.42
B_notch60	0	2	7.46	8.01	6.39	9.96
	1	3	7.55	8.31	6.55	10.41
	2	3	7.54	8.07	6.38	10.08
C_interp5060	0	2	7.46	8.03	6.39	9.99
	1	3	7.56	8.31	6.54	10.42
	2	3	7.56	8.10	6.39	10.13

```
pooled seed sd (years): val_mae 0.061, test_mae 0.138, test_normal_mae 0.100, test_abno
rmal_mae 0.194
single-pair threshold τ on TUAB test (Eq. 5.2): 0.38 years; 3-vs-3-seed threshold (Eq.
5.3): 0.22 years
```

Table 5.2. Effect of each change

One row per tested change. Columns:

- `id` : the row's id, cited in Tables 4.1, 4.2 and 5.3.
- `phase` : 1 (registry, Table 2.2), 2 (Table 2.3), L (line-removal A/B, Table 5.1), C (CV, Table 2.4).

- `baseline → variant`: the two runs compared; `bundle` marks a change of several things at once.
- `Δ val`: Eq. (0.4) on the validation set the two runs share (phase 1: TUAB val; phase 2: TUEG val v2 where both runs were scored on it).
- `Δ test`, `Δ normal`, `Δ abnormal`: Eq. (0.4) on TUAB test, overall and per cohort.
- `noise`: $\pm\tau$ (Eqs. (5.2), (5.3)) or the 95 % interval (Eq. (5.4) / bootstrap).
- `verdict`: the rule above, on `Δ test`.

id	phase	change	baseline → variant	Δ val	Δ test	Δ normal	Δ abnormal	noise	verdict
P1	1	common average reference on the input	B → D	-0.63	-0.61	–	–	± 0.38	improves
P26	1	score the first 5 min instead of whole recordings	A → B	0.11	-0.12	–	–	± 0.38	n.s.
P2	1	4 epochs / 1 warmup instead of 15 / 2	D → E2	-0.09	-0.05	–	–	± 0.38	n.s.
P3	1	mixup α 0.4	E2 → M1	-0.22	0.20	–	–	± 0.38	n.s.
P4	1	C-Mixup (σ 5 y)	E2 → M1c	-0.14	-0.02	–	–	± 0.38	n.s.
P5	1	EMA weights for evaluation	E2 → M2	-0.10	-0.08	–	–	± 0.38	n.s.
P6	1	layer decay 0.5 (stronger)	E2 → F1	0.06	0.11	–	–	± 0.38	n.s.
P7	1	layer decay 0.75 (weaker)	E2 → H1	0.03	0.42	–	–	± 0.38	hurts
P8	1	dropout 0.1	E2 → F2	5.57	6.92	–	–	± 0.38	hurts
P9	1	drop path 0.2	E2 → F3	0.02	0.16	–	–	± 0.38	n.s.
P10	1	weight decay 0.1	E2 → F4	-0.02	-0.01	–	–	± 0.38	n.s.
P11	1	head init scale 1.0	E2 → G1	0.08	0.13	–	–	± 0.38	n.s.
P12	1	LoRA r16 instead of full fine-tuning (bundle)	E2 → M3	0.00	0.99	–	–	± 0.38	hurts
P13	1	L1 loss instead of Huber	B → D2	-0.04	0.41	–	–	± 0.38	hurts
P14	1	Huber δ 2	D → DX2	0.02	0.11	–	–	± 0.38	n.s.
P15	1	soft labels (σ 2.5 y)	E2 → M4	-0.01	0.34	–	–	± 0.38	n.s.
P16	1	age-balanced loss (sqrt-inverse)	E2 → B1	0.15	0.23	–	–	± 0.38	n.s.
P17	1	codebook amplitude reconstruction	D → DX1	-0.05	0.24	–	–	± 0.38	n.s.
P18	1	codebook quantization term	D → DX4	-0.10	0.14	–	–	± 0.38	n.s.
P19	1	30 s windows instead of 10 s	B → C	-0.16	0.44	–	–	± 0.38	hurts
P20	1	random crop	D → E1	0.05	0.26	–	–	± 0.38	n.s.
P21	1	gradient clipping 1.0	D → D-clip1	-0.07	0.06	–	–	± 0.38	n.s.
P22	1	class token instead of mean pooling	D → D-cls	0.13	-0.06	–	–	± 0.38	n.s.
P23	1	[mean ; class token] concatenated	D → D-clsmean	0.03	0.05	–	–	± 0.38	n.s.
P24	1	multi-task head (age + normal/abnormal)	M1 → MT1	0.08	-0.07	0.13	-0.30	± 0.38	n.s.
P25	1	pretrained LaBraM weights instead of training from scratch	S1 → D	-1.02	-1.12	–	–	± 0.38	improves
L1	L	50 + 60 Hz interpolation instead of 50 Hz notch	A_notch50 → C_interp5060	-0.08	-0.21	-0.10	-0.34	± 0.22	n.s.
L2	L	60 Hz notch instead of 50 Hz notch	A_notch50 → B_notch60	-0.09	-0.23	-0.11	-0.37	± 0.22	improves
Q1	2	add every TUEG recording (~10x data)	Control → Exp3	–	-0.97	-0.85	-1.11	± 0.38	improves
Q2	2	train on normal recordings only (TUAB + TUEG) (bundle)	Control → Exp1	-0.62	-0.20	-0.43	0.08	± 0.38	n.s.
Q3	2	add abnormal + unlabelled recordings to normal-only	Exp1 → Exp3	–	-0.77	-0.41	-1.19	± 0.38	improves
Q4	2	TUEG normal only (no TUAB in training)	Exp1 → Exp2	0.07	-0.03	0.07	-0.16	± 0.38	n.s.
Q5	2	10 / 3 epochs on the normal cohort	Exp1 → Exp1b	-0.02	0.08	0.11	0.05	± 0.38	n.s.
Q6	2	10 / 3 epochs on the all cohort	Exp3 → Exp3b	–	0.43	0.60	0.23	± 0.38	hurts
Q7	2	window rejection (±800 µV, flat channel)	Exp3 → Exp3q	–	0.07	0.12	0.03	± 0.38	n.s.
Q8	2	input clip ±800 µV	Exp3 → Exp3c	–	0.02	0.09	-0.06	± 0.38	n.s.
Q9	2	classical features as transformer tokens	Exp1 → Exp1t	–	0.76	1.19	0.26	± 0.38	hurts
Q10	2	classical features only, no EEG encoder	Exp1 → Exp1f	–	0.10	0.22	-0.06	± 0.38	n.s.
Q11	2	shared TUEG patient split (v2)	Exp3 → Exp3v2	–	0.15	-0.11	0.47	± 0.38	n.s.
Q12	2	status + pathology condition vectors	Exp3v2 → Exp3cond	-0.10	-0.14	-0.07	-0.22	± 0.38	n.s.
Q13	2	class token (with conditions)	Exp3cond → Exp3clscond	0.07	-0.03	-0.03	-0.02	± 0.38	n.s.
Q14	2	cohort v3 + missing-T1/T2 handling + speed settings (bundle)	Exp3v2 → Exp4base	-0.14	-0.05	0.16	-0.31	± 0.38	n.s.
Q15	2	compact copy of the cohort (reproduction)	Exp4base → Exp4basec	-0.01	0.06	-0.10	0.24	± 0.38	n.s.
Q16	2	class token + conditions + 10 ep with early stopping (bundle)	Exp4base → Exp4ctrl	0.05	-0.06	-0.27	0.19	± 0.38	n.s.
Q17	2	T1/T2 dropout + consistency (6 vs 10 epochs) (bundle)	Exp4ctrl → Exp4aug	-0.07	-0.01	0.09	-0.14	± 0.38	n.s.
Q18	2	T1/T2 dropped everywhere (21 channels) + cohort v3 (bundle)	Exp3cond → Exp4drop	-0.15	-0.00	-0.13	0.16	± 0.38	n.s.
Q19	2	15 / 5 epochs with early stopping (aug)	Exp4aug → Exp4aug15	0.06	-0.12	-0.06	-0.20	± 0.38	n.s.
C1	C	class token + conditions + 10 / 3 epochs (CV) (bundle)	cvbase → cvctrl	-0.07	-0.21	-0.28	-0.13	[-0.41, -0.01], better in 5/5 folds	improves
C2	C	T1/T2 dropout + consistency λ 0.1 (CV)	cvctrl → cvaug	-0.03	0.04	0.02	0.07	[-0.16, +0.25], better in 2/5 folds	n.s.
C3	C	median of 5 fold models instead of one (cvctrl)	cvctrl single → cvctrl ensemble	–	-0.18	-0.22	-0.13	[-0.27, -0.09] (bootstrap)	improves
C3b	C	median of 5 fold models instead of one (cvbase)	cvbase single → cvbase ensemble	–	-0.05	-0.06	-0.04	[-0.11, +0.01] (bootstrap)	n.s.

Figure 5.1. Effect sizes

What is plotted. One row per tested change (Table 5.2). The point is how much that change moved the TUAB test MAE, Eq. (0.4):

$$\Delta\mathrm{MAE} = \mathrm{MAE}(\text{run with the change}) - \mathrm{MAE}(\text{run without it}),$$

in years, on all 274 test recordings. The two runs are named in brackets in the row label (baseline → variant).

This is a **difference of two errors, not a prediction error**. A point at −0.5 means the change made the model 0.5 years more accurate. It does not mean the model predicts ages 0.5 years too low.

- **Left of 0:** the change lowered the error, so it helped. Further left is better.
- **At 0:** no effect.
- **Right of 0:** the change raised the error, so it hurt.

Bars and colours.

- The horizontal bar is the noise band: how far two runs differing only by chance would land apart. It is $\pm\tau$ (Eq. (5.2)) for single runs and 3-seed pairs, and the 95 % interval (Eq. (5.4) or the bootstrap) for CV and ensemble rows.
- **Green (`improves`):** the whole band lies left of 0, so the gain is larger than seed noise.
- **Red (`hurts`):** the whole band lies right of 0.
- **Grey (`n.s.`):** the band crosses 0. The effect cannot be told apart from noise.

Changes beyond ±2 years (collapsed runs) are drawn at the edge, with their value written next to them. Panels:

- (a) Phase 1, TUAB-only runs.
- (b) Line removal, phase 2 and CV.

The signed drift of the predictions themselves, mean(predicted − true), where closer to 0 is better, is the bias of Eq. (0.3). It is shown per model and status in Table 3.4 (`bias`) and in Figure 3.1.

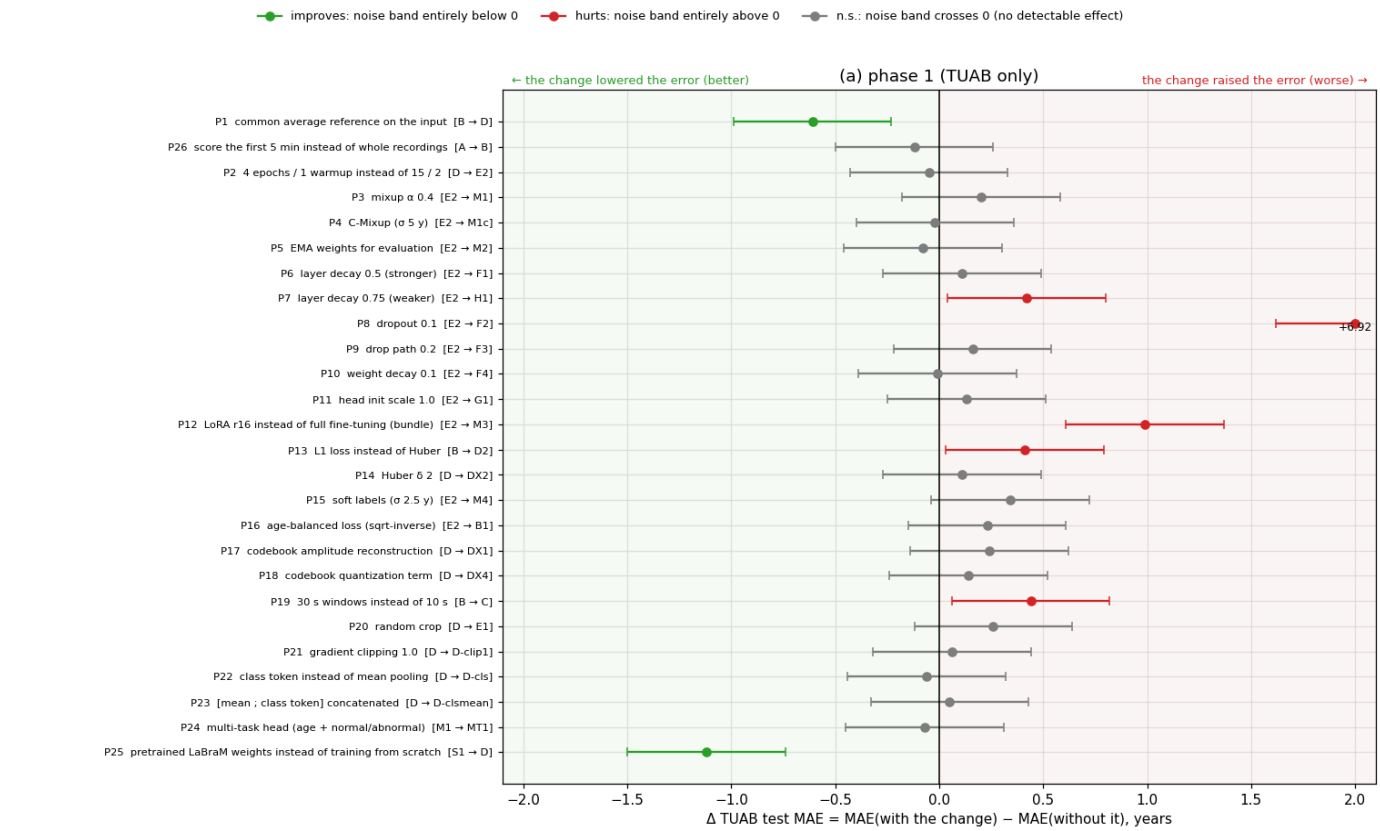


Table 5.3. Changes versus LaBraM: which mattered

The summary the study was after. One row per change from the LaBraM TUAB recipe to the best configuration (Tables 4.1 and 4.2), with:

- `estimate`: the Table 5.2 row that isolates the change best and its Δ test;

- `verdict` : that row's verdict;
- `in best config` : whether the change is part of `cvctrl` .

`not isolated` means no run changes it alone, so its effect is not known.

change vs LaBraM	estimate	Δ test	verdict	in best config
training data: TUAB → TUAB + TUEG (~10x recordings)	Q1: Control → Exp3	-0.97	improves	yes
abnormal and unlabelled recordings in training	Q3: Exp1 → Exp3	-0.77	improves	yes
pretrained LaBraM weights (vs from scratch)	P25: S1 → D	-1.12	improves	yes
common average reference on the input	P1: B → D	-0.61	improves	yes
mains: 50 + 60 Hz interpolation instead of 50 Hz notch	L1: A_notch50 → C_interp5060	-0.21	n.s.	yes
ensemble: median of the 5 fold models	C3: cvctrl single → cvctrl ensemble	-0.18	improves	yes
class token + condition vectors + 10 epochs (bundle)	C1: cvbase → cvctrl	-0.21	improves	yes
condition vectors alone (single split)	Q12: Exp3v2 → Exp3cond	-0.14	n.s.	yes
class-token pooling alone (TUAB / TUH)	P22: D → D-cls	-0.06	n.s.	yes
short schedule (4–10 epochs instead of 15–50)	P2: D → E2	-0.05	n.s.	yes
mixup α 0.4	P3: E2 → M1	0.20	n.s.	yes
cohort v3 + missing-T1/T2 handling + speed settings (bundle)	Q14: Exp3v2 → Exp4base	-0.05	n.s.	yes
5-minute evaluation window	P26: A → B	-0.12	n.s.	yes
T1/T2 channel-dropout augmentation	C2: cvctrl → cvaug	0.04	n.s.	no (cvaug)
longer schedule on the full cohort (10 epochs)	Q6: Exp3 → Exp3b	0.43	hurts	no
dropout 0.1	P8: E2 → F2	6.92	hurts	no
weaker layer decay 0.75	P7: E2 → H1	0.42	hurts	no
LoRA instead of full fine-tuning	P12: E2 → M3	0.99	hurts	no
30 s windows	P19: B → C	0.44	hurts	no
classical features as tokens	Q9: Exp1 → Exp1t	0.76	hurts	no
L1 loss	P13: B → D2	0.41	hurts	no
weight decay, drop path, layer decay 0.5, head init	P10: E2 → F4	-0.01	n.s.	no
loss shape: Huber δ 2, soft labels, age-balanced	P16: E2 → B1	0.23	n.s.	no
codebook regularization	P18: D → DX4	0.14	n.s.	no
signal QC: window rejection / ±800 µV clip	Q7: Exp3 → Exp3q	0.07	n.s.	no
multi-task head	P24: M1 → MT1	-0.07	n.s.	no
trim 60 s at both ends; lr 1e-4 (vs 5e-4); batch 128	–	–	not isolated	yes

6. Conclusions

Generated from the tables above, so they stay in step with the data when the notebook is re-run.

- **Noise floor.** Seed-to-seed sd of the TUAB test MAE is 0.14 years (Table 5.1), so two single runs must differ by more than 0.38 years to count (Eq. (5.2)). Of 51 tested changes, 7 improve, 7 hurt and 37 are within noise (Table 5.2).
- **What improved the error (Table 5.2):** P25 pretrained LaBraM weights instead of training from scratch (-1.12); Q1 add every TUEG recording (~10x data) (-0.97); Q3 add abnormal + unlabelled recordings to normal-only (-0.77); P1 common average reference on the input (-0.61); L2 60 Hz notch instead of 50 Hz notch (-0.23); C1 class token + conditions + 10 / 3 epochs (CV) (bundle) (-0.21); C3 median of 5 fold models instead of one (cvctrl) (-0.18).
- **What hurt:** P13 L1 loss instead of Huber (+0.41); P7 layer decay 0.75 (weaker) (+0.42); Q6 10 / 3 epochs on the all cohort (+0.43); P19 30 s windows instead of 10 s (+0.44); Q9 classical features as transformer tokens (+0.76); P12 LoRA r16 instead of full fine-tuning (bundle) (+0.99); P8 dropout 0.1 (+6.92).
- **Best configuration: `cvctrl`.** TUAB test MAE 7.28 ± 0.14 per fold model (normal 5.53, abnormal 9.37; Table 3.3). Its head is given the status/pathology vector.
- **Best EEG-only: `cvbase`.** 7.49 ± 0.09 per fold model, normal recordings 5.81, against 6.60 for the best published TUAB normal result (Gemein et al. 2024, different split and training data).
- **Ensembles.** The median of the 5 fold models scores 7.10 (`cvctrl`) and 7.44 (`cvbase`) on the TUAB test, normal 5.75 (`cvbase`).
- **Most of the gain over LaBraM comes from data.** The largest effects are:
  - more and more varied training recordings (Q1, Q3);
  - pretrained weights (P25);
  - the common average reference (P1). Architecture, loss and regularization changes are within noise or harmful (Table 5.3).
- **The abnormal cohort carries the error.** Abnormal recordings stay about 4 years worse than normal ones in every configuration (Tables 2.3, 3.3).

Caveats.

- **Selection on test:** phases 1 and 2 chose among configurations on the TUAB test set, so their best single-split numbers are optimistic. The CV and TUEG out-of-fold numbers are not.
- **Bundles:** C1, Q14, Q16, Q17 and Q18 change several things at once, so their parts are not separately known.
- **Condition vectors:** `cvctrl` and `cvaug` use the status/pathology vector at test time. Only `cvbase` is comparable with EEG-only brain-age models.
- **Noise floor scope:** it comes from one configuration, the E2 recipe on TUAB (Table 5.1), and is assumed to transfer to the larger cohorts.